

Hidden in the Dust:
Searching for Active Galactic Nuclei in JWST's
COSMOS-Web Field

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Abstract

Imaging from JWST/NIRCam has revealed dust-obscured [active galactic nuclei \(AGN\)](#) with a much higher frequency than previously seen with the rest-frame UV and optical imaging done by the HST. Moreover, the frequency of AGN is higher in the longer wavelength JWST/NIRCam bands such as the F444W. In this study we quantify the frequency of such highly obscured AGN among high redshift galaxies identified as hosts of an AGN using near-infrared, radio, and x-ray tracers. A specific goal of our work was to analyze the morphology of a population of faint radio sources in the COSMOS field in a JWST/NIRCam band and establish the frequency of highly obscured AGN among them. We fit all 727 galaxies with a double-Sérsic profile and a Sérsic profile plus a point-source component. We compare the fractional flux percentages of both models to identify the 256 galaxies in our sample with unresolved point sources. Our work expands on that of Ortiz et al. 2024, by analyzing a sample that is 11 times larger. Our method of morphological analysis can be used to search more broadly for unresolved infrared-luminous AGN. This could help answer questions about the evolution of galaxies, including our own.

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Chapter 1

Introduction

“For a moment, nothing happened. Then, after a second or so, nothing continued to happen.”

Douglas Adams, *Hitchhiker’s Guide to the Galaxy*

A hundred years ago, Edwin Hubble discovered an object that would change how we understand the cosmos: NGC 6822, the first object definitively attributed to a region outside the galactic system (Hubble, 1925). This object is a faint cluster of stars and nebulae that we now consider a barred irregular galaxy. Hubble was able to identify 11 Cepheid variable stars within this system as seen in 1.1. These are stars that vary on a consistent timescale such that we are able to calculate distance based on their luminosity. In the case of NGC 6822, Hubble measured the distances to be 214,000 parsecs, far outside the Milky Way. The distance turned out to be wrong, though he was right about NGC 6822’s extragalactic nature. Hubble made an incorrect assumption about the calibration of the periodicity-luminosity relation. We now know that NGC 6822 is a distance of 490,560 pc away from us, over double the distance Hubble estimated in 1925.

This discovery led to a reconsideration of Messier objects like M31, the Andromeda Galaxy. There could be galaxies other than our own. Hubble (1929) began studying the structure and size of M31, which has a distinct spiral structure and an unresolved nuclear region. This research expanded the nebulae

classification that Hubble developed in [Hubble \(1926\)](#). These original classifications are irregular, elliptical, and spiral. Irregular galaxies have no noticeable symmetry or central nucleus. Elliptical galaxies are smooth, with featureless distributions of light and complete rotational symmetry. Spiral galaxies have a flattened disk with stars forming along spiral arms and a central nucleus called the bulge. He also identified Lenticular galaxies that do not have spiral arms but do have the disk and bulge structure of spiral galaxies. Hubble's discoveries advanced our thinking about the big questions about our place in the universe: How did we get here? How did the structure of the Milky Way Develop? What is the structure of the Universe? In working to understand these questions, we as astronomers have investigated topics that span multiple scientific disciplines, from astrobiology to extra-galactic astronomy.

As the study of the newly discovered galaxies grew, astronomers began noticing some unique features that led to the discovery of [active galactic nuclei \(AGN\)](#). Carl Keenan Seyfert ([Seyfert, 1943](#)), initially identified them as galaxies with broad nuclear emission lines. These Seyfert galaxies, as they became known, are one of the main broad classifications of AGN today. At the time, though, research into these objects was not astronomers' main focus. Further investigation into AGN would have to wait for the development of radio astronomy ([Shields, 1999](#)).

It was 1960 when astronomers discovered quasars as a subset of radio galaxies. At the December meeting of the American Astronomical Society, Allan Sandage of the Mount Wilson and Palomar Observatories reported his research into 3C 48, which, due to its small radio diameter, was generally considered to be a nearby star in our galaxy with odd properties ([Greenstein, 1963](#); [Shields, 1999](#)). After this discovery, Sandage and Maarten Schmidt of the California Institute of Technology (Caltech) worked with Thomas A. Matthews to use the new interferometer at the Owens Valley Radio Observatory operated by Caltech to begin to identify more odd quasi-stellar objects. These objects had odd, unrecognizable spectra and studying these objects would spark a greater understanding of the structure and form of AGN.

Two years later, in 1962, Schmidt obtained the spectra of quasar 3C 273, which contained broad emission lines that were difficult to identify. This object was clearly no ordinary star. The comparison between these two quasars, 3C 273 and 3C 38, revealed that the spectra looked similar but their broad emission lines were at different wavelengths ([Greenstein & Schmidt, 1964](#)). Schmidt concluded this was only possible because the emission lines had been redshifted

to longer wavelengths (Schmidt, 1963). This insight allowed him to identify the emission lines as the Balmer series of Hydrogen: a pattern of decreasing strength and spacing toward bluer light. With a redshift of $z = 0.16$ Schmidt was able to match the emission lines to the $H\beta$, $H\gamma$, $H\delta$, and $H\epsilon$ transitions of the Balmer series (Shields, 1999).

The identification of the spectrum of 3C 273 led Schmidt to classify it as a stellar object. He offered two possible explanations in his 1963 paper. The first, that 3C 273 is a stellar object with a large gravitational redshift. In this scenario, its radius would have to be on the order of 10 km. Schmidt acknowledges that the occurrence of permitted lines (emission lines found on Earth) and a forbidden line (emission lines not found on Earth) at the same redshift would be extremely unlikely. The second possibility (and what we now know to be correct) is that the "stellar" object lies within the nuclear region of a galaxy with a cosmological redshift of 0.158. This means that the diameter of the region would have to be less than 1 kpc and a hundred times optically brighter than the luminous radio galaxies identified thus far. Solving the mystery of 3C 273 led to a new understanding of Greenstein's spectrum of 3C 48. It was less likely a nearby star in our galaxy and more likely the nuclear region of a galaxy (Greenstein, 1963). The posited redshift of 3C 273 allowed the identification of Mg II lines within the spectrum (Schmidt, 1963). Following this, a redshift of $z = 0.3675$ was posited for 3C 48 which was supported by the presence of Mg II lines in this object as well (Greenstein & Schmidt, 1964; Shields, 1999).

While observing targets like 3C 273, Sandage noted that their brightness changes over time, although he was not certain if this was periodic or random (Sandage, 1964).

Upon the discovery of 3C 48, Sandage and Matthews took three-color photoelectric observations on the UBV system of the target. (Matthews & Sandage, 1963). They found a 6-month optical-flux variation, as seen in Fig 1.3. But their initial research into the source of the variation was inconclusive. Later studies revealed a redshift of $z = 0.3675$ that implied a size limit of less than 0.15 pc on the object. This, combined with the redshift, suggests that 3C 48 exists as its own galaxy, 1100 Mpc away. This class of objects became known as quasi-stellar galaxies and are now called quasars.

A 1965 paper, (Sandage, 1965), found that quasars, compared to normal stars, showed an "ultraviolet excess" when compared on a color-color diagram. As

more data was collected on objects with strong UV continua, Sandage noticed that the number of objects did not match with known radio sources. These targets, [quasi-stellar galaxies \(QSG\)](#), were thought to have high redshifts on the order of $z = 1.241$. Due to their high redshifts these QSGs were potentially very promising for studying cosmological questions about galaxy formation and the particulars of the Milky Way’s evolution.

In the late 1970s to early 1980s a survey of the optical and [infrared \(IR\)](#) energy distribution of QSOs revealed broad bumps around $3\mu\text{m}$ that was theorized to be from dust emission. Further measurements at longer and shorter wavelengths by the *International Ultraviolet Explorer* and the *Infrared Astronomical Satellite* as well as ground based telescopes showed that the 3-5 μm “bump” was present in most Seyfert galaxies and QSOs ([Shields, 1999](#)).

Seyfert Type 1 galaxies without reddening appeared to need a hot thermal component. The most popular explanation was emissions from an accretion disk. Research to understand the structure of disks—especially that of Shakura & Sunyaev—concluded that disks were geometrically thin and the vertical and radial directions could be analyzed separately. [Shakura & Sunyaev \(1973\)](#) characterized disks with dimensionless component α that split the disk into 3 radial sections based on the relative importance of radiation pressure, gas pressure, electron scattering and absorption opacity. The most power was produced by the innermost region, dominated by electron scattering and radiation pressure.

Alongside the discoveries of QSOs and measurements in the IR and UV in the 1960s, the *Uhuru* satellite detected X-rays from Seyfert galaxies. The satellite also revealed, among other things, binary systems involving black holes. When a system with a black hole was detected, the x-rays were attributed to the infalling matter from the companion star as it spiraled into the black hole by way of a disk ([Shields, 1999](#)). The scale of AGNs might be bigger, but their disks also consist of infalling gas and dust.

As astronomers discovered more about quasars and the structure of AGN they developed a unified model for AGN structure, which we discuss in [2.1.5](#). But one of the biggest questions about AGN evolution remains: How does the structure develop alongside galaxies? ([Netzer, 2015](#)).

Our work could help identify new populations of AGN, specifically radio-quiet AGN. That could shine light onto the mysterious Little Red Dots, which have

been found to be mostly radio silent (Perger et al., 2025) and could play a role in galaxy formation. And it could advance our understanding of how closely AGN formation is linked to galaxy formation in the early universe.

This work assumes a Λ CDM universe. There are three major components of this model: Λ is a cosmological constant used to describe dark energy, the unknown source of energy driving the expansion of the universe; the theorized cold dark matter (CDM); and ordinary matter. Cold dark matter is an unknown form of matter that does not interact with the electromagnetic field. We know of its existence through the gravitational force it exerts on its surroundings. The main parameters of the Λ CDM model are: the Hubble constant written as H_0 which is current rate of the expansion of the universe; the curvature parameter, $\Omega_0 = 1$, in this case it is set to a flat universe; the density parameters, which describe the amount of energy divided by the volume. The density is not one parameter but rather several that describe the division of energy, from radiation to baryonic matter to dark matter.

Part 1 of this work has two chapters. In the first we provide background information. We discuss the development of AGN in the early universe and possible formation methods of massive and supermassive black holes, as well as the unified model of AGN and how it ties into structure and categorization of AGN. The second chapter explains the coding packages utilized in this work and discusses the morphological fitting methods performed. We will also discuss the selection methods for our data.

Part 2 of this work has three chapters. In the first we relay the main analytical results from the morphological fitting and analysis. The second contains our discussion, comparing this work’s conclusions to that of Ortiz et al. (2024), as well as considering some of the challenges and the greater significance of our results. In the third and final chapter, we summarize our conclusions, discuss limitations of our current methods, and describe possible future work that expands on the results we’ve achieved here.

Part 3 contains our two appendices. The first has other non-central explorations of the data that we undertook as well as additional justification for some of the choices we made in this work. The second has information about accessing the code and data used in or produced by this work as well as snippets of each. This is followed by a glossary of terms and a list of acronyms.

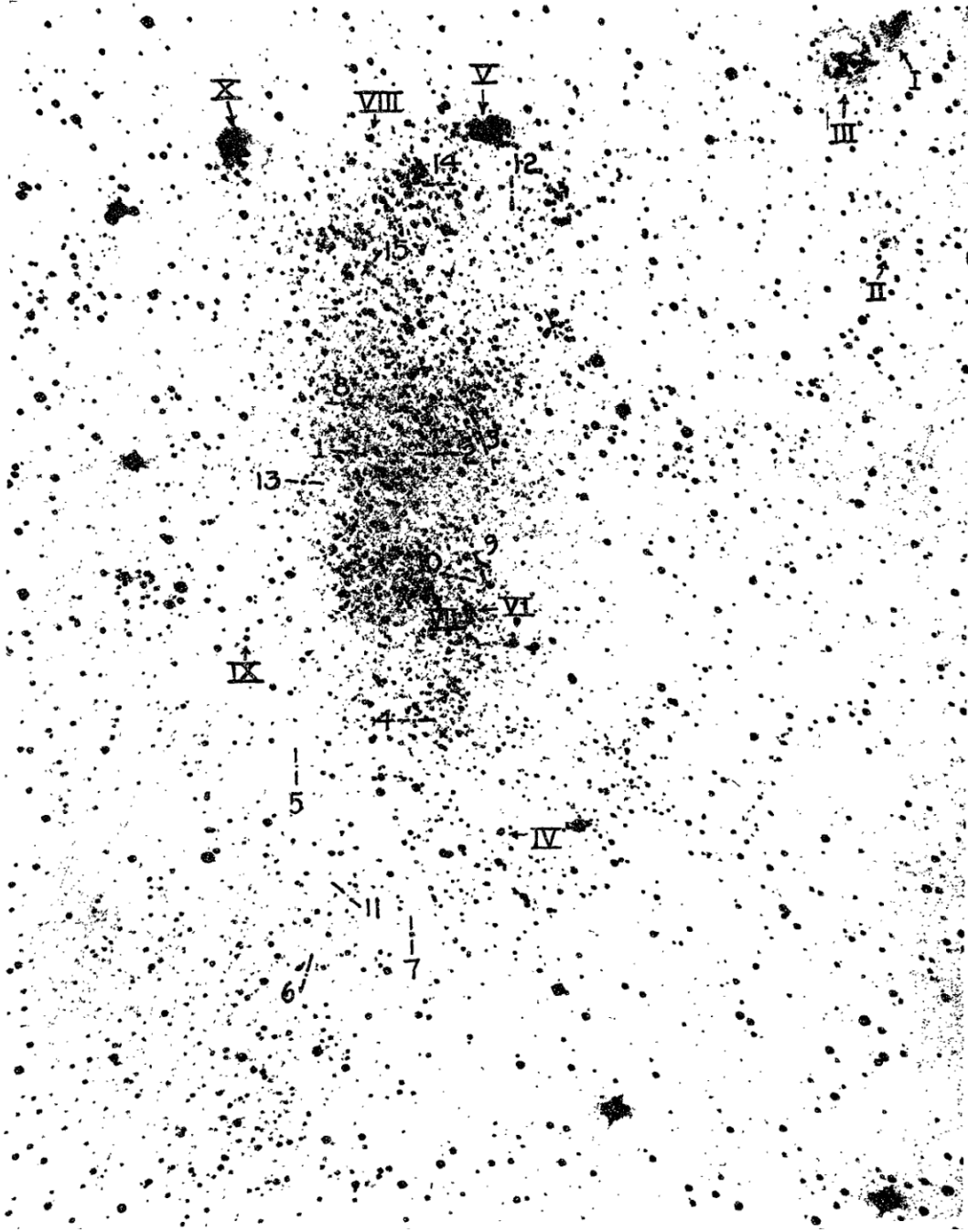


Figure 1.1 A figure from (Hubble, 1925) showing a negative print image taken of NGC6822. The variable stars identified in the galaxy are labeled with Arabic numerals; the nebulae involved in or superimposed on the galaxy are identified with roman numerals.

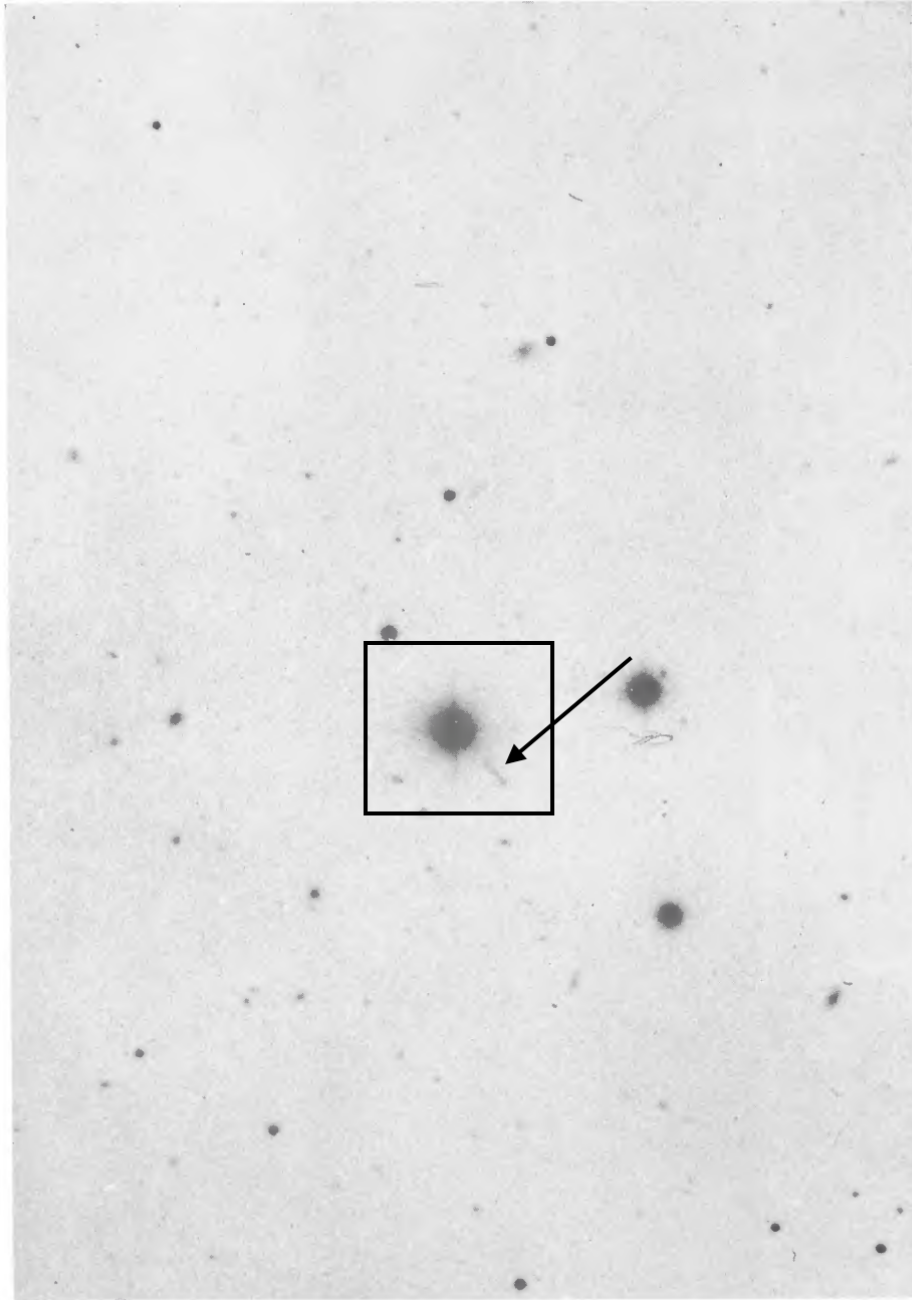


Figure 1.2 Enlarged photo of 3C 273 taken by Sandage. There is a weak narrow jet visible at $\text{pa } 223^\circ$ that reaches about $20''$ from the [quasi-stellar objects \(QSO\)](#). The QSO and jet are labeled with a box and arrow respectively. Identifying a radio source that matched so well with an optical object helped Sandage posit that the object's spectrum could be understood with its high redshift. Figure 1 from [Greenstein & Schmidt \(1964\)](#).

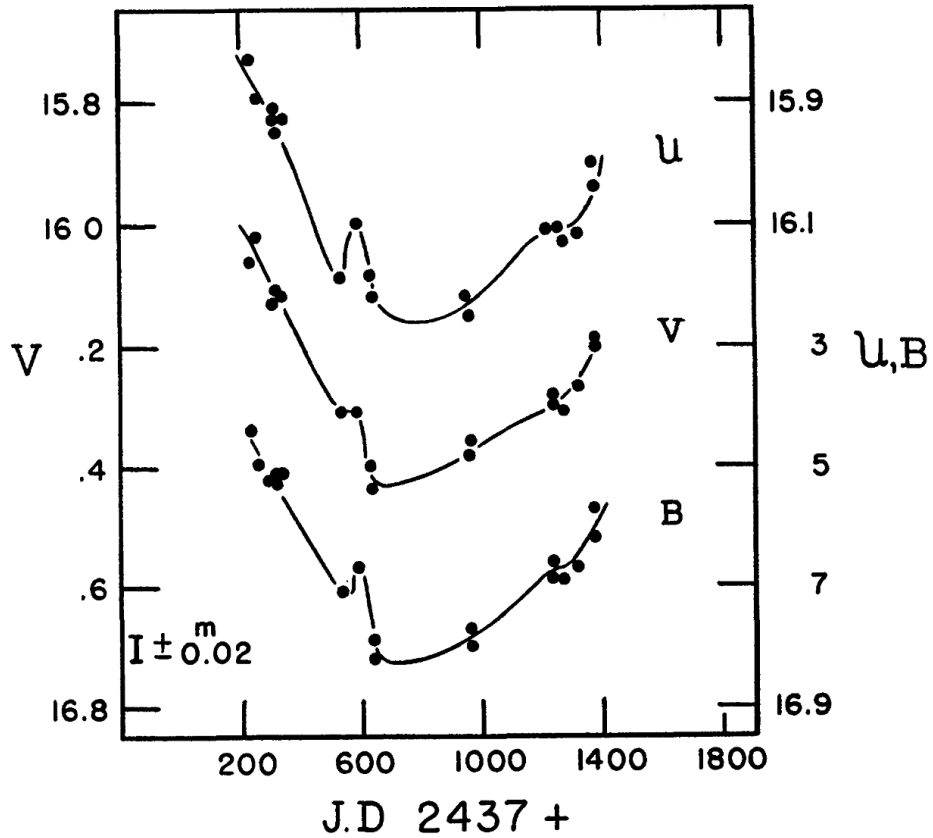


Figure 1.3 A figure from Sandage (1964) showing the intensity of light from 3C 48 over time in three different bands. Note the difference in scales for the V versus the U and B bands. There is variation both long and short timescales. The largest magnitude jump measured by Sandage is 1.4.

Part I

Background and Methods

Chapter 2

Background

The universe began infinitesimally small and extremely hot. Then, eventually, came the Big Bang, the moment everything changed. It was not exactly a bang but a very short period of rapid inflation. Within the span of 10^{-35} seconds the universe grew by a factor of 10^{26} . This inflation sets in motion everything that follows. All the matter that before was tightly packed has now spread into a nearly homogeneous hot soup of fundamental particles. This rapid expansion still leaves the universe nowhere near the size it is today. The observable universe, today 96 billion light years across, was just a meter wide. At this point the universe was too hot to form even protons or neutrons. But as the expansion continues, the soup is able to cool and form the very first protons and neutrons in the universe. As the expansion overtakes the rate of these weak interactions, the fundamental particles are now fixed in place. All this in just one second. Small perturbations in the plasma of the universe produce under- and over-densities within the plasma that will be critical as development continues.

Over the next few minutes, the existing supply of protons, neutrons, and electrons attempt to form atoms. It is still too hot for the electrons to join up, but the neutrons and protons can form deuterium and later tritium, hydrogen isotopes with one and two neutrons respectively. This process is called nucleosynthesis and still occurs, at times, in our universe today under extreme conditions. Eventually, the universe cools to the point that such nucleosynthesis was no longer possible, but electrons were still not able to join up with the groups of protons and neutrons. It would take 380,000 years for the soup

of ionized particles to cool enough that electrons could attach and form atoms (mostly hydrogen and helium), in a process called recombination. This process made the universe transparent. Before recombination, all the electrons flying around meant that light couldn't get very far before running into something, which meant the universe was completely opaque (as if standing in the middle of star). But with electrons settling down in the cooling universe, the light could travel uninterrupted. This light forms what astronomers call the Cosmic Microwave Background or CMB, low level radiation that permeates the universe, preserving a snapshot of the early universe.

We've been discussing baryonic matter, but that's not all there is. Dark matter has been lurking the whole time, and those quantum fluctuations in density we mentioned earlier begin to grow over time as gravity draws in more dark matter. This eventually collapses and forms a dark matter halo. This in turn forms a network of filaments and nodes across the entire universe, as seen in Figure 2.1. This pattern is what we call the "cosmic web" today. At this point, the baryonic matter is beginning to cluster in the center of these dark matter halos and, given some time (a few hundred million years), they eventually form the first stars and galaxies of the universe. This creates the large scale structure of the universe.

The creation of stars and galaxies introduces the possibility for massive stars to explode and form the first stellar mass black holes. But the supermassive black holes that power AGN are much higher mass than a stellar black hole, often a mass upwards of 10^6 Solar masses. This means a standard stellar mass black hole is likely not the source of these powerful celestial objects. This brings into question by what methods these supermassive black holes grew, and we discuss a few of the most likely possibilities in 2.0.1.

2.0.1 Black Hole Seeds

To understand how massive black holes formed in the early universe, we need to understand what the black hole seed population looked like in the first hundreds of millions of years of the universe. A black hole seed is defined by Volonteri et al. (2021) as a black hole of yet unconstrained initial mass within a range of a few hundred to on the order of a million solar masses. Those on the lower end of the spectrum are considered "light" seed while those on the upper end are considered "heavy." The easiest and simplest case assumes all black holes originated from the death of massive stars. But this doesn't explain how a black hole on the order of 10^3 stellar masses would reach the

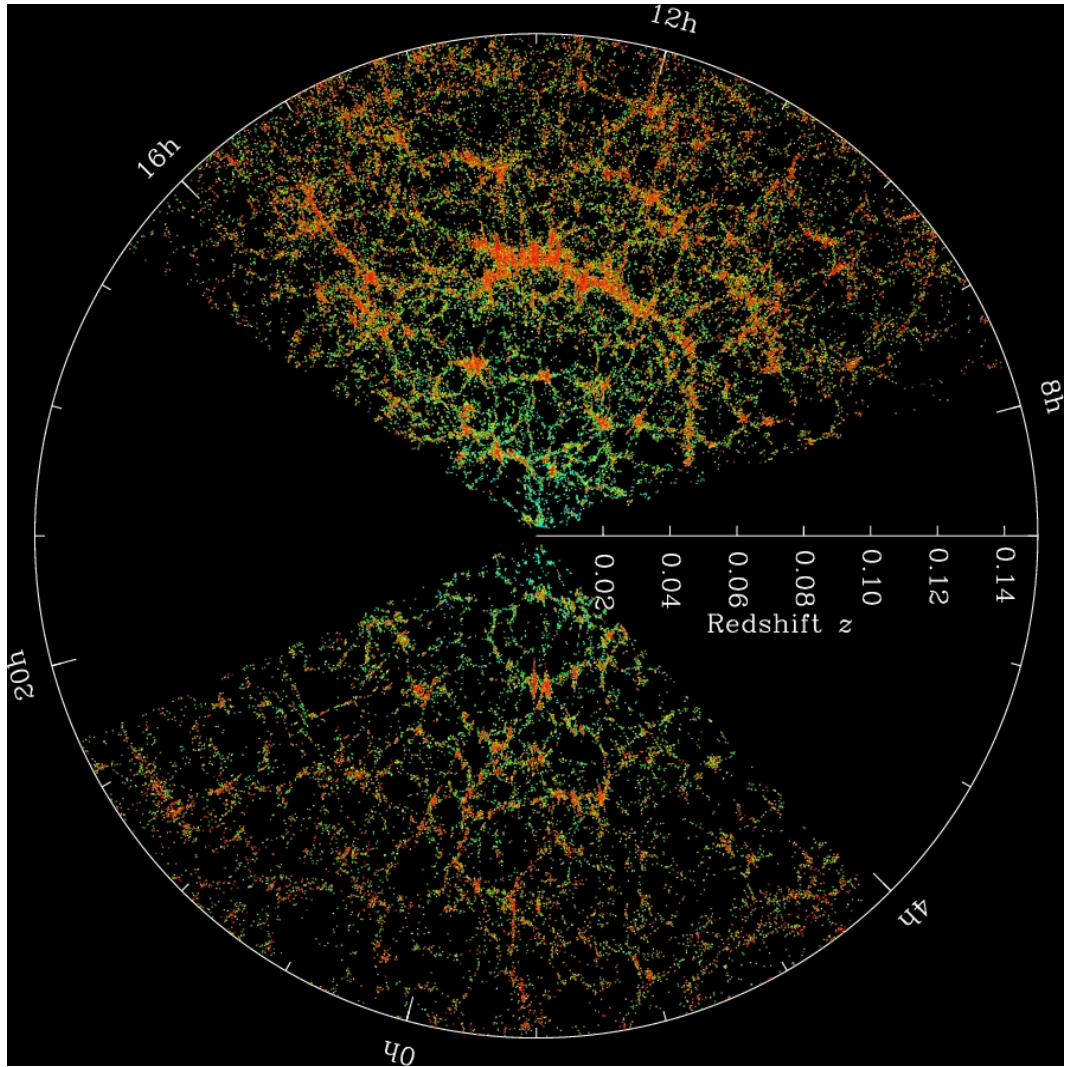


Figure 2.1 Plotting of galaxy density in the universe from the Sloan Digital Sky Survey. Galaxies cluster in spots of over-density, where there is more dark matter present.

$10^5 \leq$ massive black holes we see today. Heavy seeds are thought to be the result of direct collapse of massive gas clouds into supermassive stars. But the conditions necessary for such an event are much rarer in the early universe than the conditions necessary for light seeds to form, as seen in Figure 2.2.

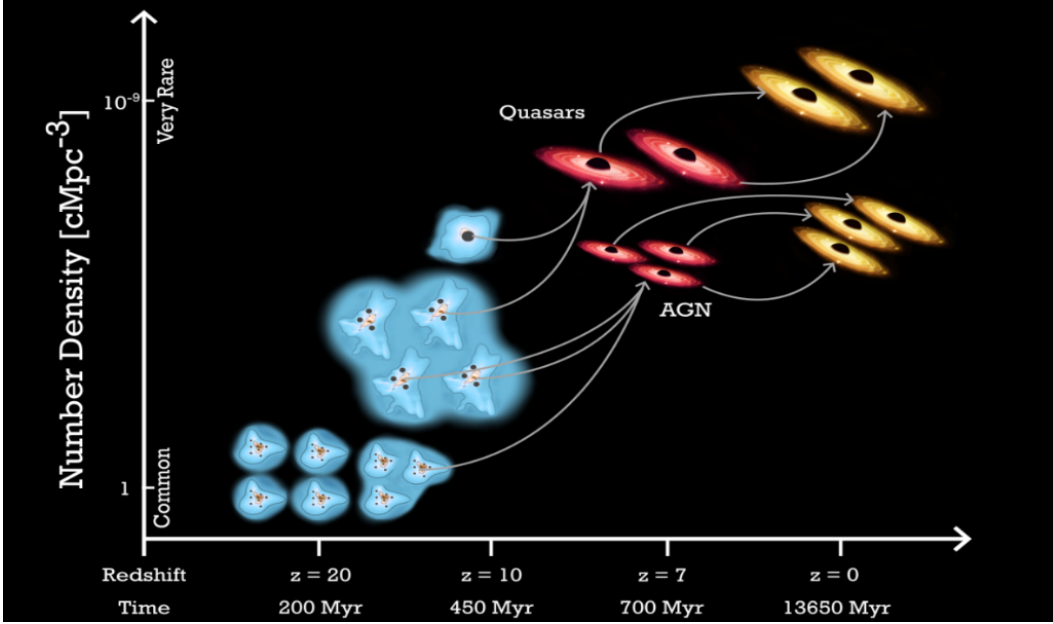


Figure 2.2 This figure depicts the rarity of black hole seed formation in the early universe. The y-axis increases in rarity while the x-axis increases in time towards a redshift of 0 (the modern day). Up until a redshift of $z = 10$ light seeds can form with relative ease but are less likely to be the source of massive high redshift AGN. Heavier seeds, however, can more easily form the observed AGN but are far less common. Figure 1 from [Regan & Volonteri \(2024\)](#).

2.0.2 Black Hole Seed Formation

Several models of heavy black seed formation have been developed over the years. [Regan & Volonteri \(2024\)](#) discuss three of the main ones: direct collapse of a supermassive star into a massive black hole, dynamical runaway processes, and direct formation of a massive black hole from galactic mergers. Runaway collisions of stars in dense clusters has been shown to form [massive black holes \(MBH\)](#), although results are very dependent on the initial conditions of the cluster. The more successful simulations were able to grow black hole seeds to tens of thousands of solar masses when including both stellar collisions on top of gas accretion rates. We discuss growth through gas accretion in more depth

in section 2.0.4. This method, while uncertain and not necessarily convergent across simulations, has shown a possible route to early heavy black hole seed formation.

On paper, it is also possible to grow an initial black hole seed to above $1000 M_{\odot}$ through hierarchical black hole (BH) mergers within a star cluster undergoing core collapse (Regan & Volonteri, 2024). In this case, one heavy black hole seed is not sufficient; rather a series of mergers of lighter black hole seeds is necessary to grow a massive black hole. Once again, this method is highly dependent on the initial conditions of the cluster.

There is also the possibility for dynamical pathways to heavy seeds through very massive or super massive stars. A supermassive star is defined by a very high accretion rate onto the stellar surface approximately more than $.001 M_{\odot}/yr$ which causes the outer shell of gas around the star to expand. These theoretical stars would be the perfect candidate for heavy black hole seeds with masses above $10^5 M_{\odot}$ and relatively cool with effective temperatures around 5000 K (Regan & Volonteri, 2024).

One final method of heavy seed formation was proposed by Mayer et al. (2024), a method that would bypass many of the other issues and create some of the most massive black holes. This theory proposes that formation of a massive black hole could be done within the merger of massive galaxies, resulting in the creation of supermassive black holes with masses up to $10^8 M_{\odot}$ (Mayer et al., 2024). The merger of these massive galaxies' halos would drive large gas inflows to create the central MBH (Regan & Volonteri, 2024). This method is potentially limited, though, because it fails to explain the existence of less massive black holes at high redshifts.

All of the possible methods of forming black hole seeds (light and heavy) are assumed to have an accretion rate that drops off after 10^5 years as the readily available gas in the halo is consumed. Gas replenishment is needed for continued growth. Gas can come from mergers with gas-rich halos or accretion from the cosmic web (Regan & Volonteri, 2024). The paper concludes that heavy seed production is more likely the more symmetric the halo is. That while both heavy and light black hole seeds are limited by their Bondi radii, heavy seeds have an easier time scaling up due to initial mass advantages. Rapid growth, however, is still challenging because it's only possible if a seed can "sink" to the center of the galaxy, where the densest population of gas lies. This favors heavier seeds; light seeds, without the mass or gravitational

influence to reach high accretion rates, tend to be distributed inhomogeneously around the halo center.

2.0.3 Black Hole Seed Growth

All the models of black hole seed formation assume accretion rate drops off when the gas in the halo is depleted. The timescales depends on outside factors that can possibly disrupt the gas. These disruptions can stem from the MBH itself or from its host galaxy. AGN have the power to emit energy on a scale that can impact processes throughout the entire galaxy, which we discuss in 2.1.3. AGN jet are a particularly dramatic example (see Figure 2.5 or 2.7), with the ability to release enough energy to sometimes heat, sometimes deplete—and sometimes heat and deplete—the gas reservoir feeding the growth of the black hole seed. This prevents further gas infall (Volonteri et al., 2021). This display is one of many forms of AGN feedback that’s considered “negative” because it suppresses energy. Feedback most commonly occurs at high accretion rates, as we discuss in Section 2.1.3. This means MBHs are going to eventually cut off their own source of gas and eventually accretion will slow until feedback has dissipated. Then this cycle will begin again (Volonteri et al., 2021).

MBHs are not the only threat to their gas reservoir. Supernovae and supernovae-driven winds, which can reach velocities of more than 200 km/s in the interstellar medium, have the potential to disrupt gas supply (Volonteri et al., 2021). At these velocities the wind can also clear dense gas from lower mass galaxies if the gravitational wells are shallow. This can mean that, until a galaxy has the mass to retain its gas, the MBHs growth could be greatly slowed or frozen. Volonteri et al. (2021) describes this as one of the main obstacles to black hole seed growth in low-mass galaxies.

MBHs accrete most efficiently in the center of galaxies with the densest population of gas. Not many forces can dislodge MBHs from this advantageous position, but if they are—or if they were too light to reach there in the first place—MBHs often struggle to grow. A MBH could be knocked out of the center after a merger with another MBH and galaxy, although this greatly depends on the mass ratio of the merger, as well as the spins, magnitude, and relative direction.

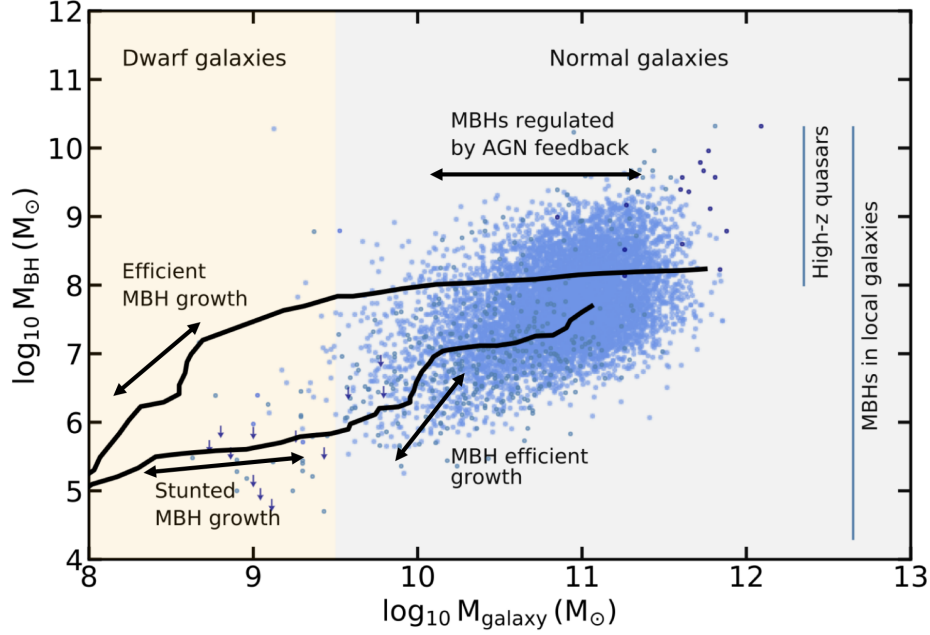


Figure 2.3 Figure 1 from [Volonteri et al. \(2021\)](#). This graph shows the growth of massive black holes in galaxies versus galaxy mass in solar masses. The blue cloud represents observational data while the black curves are from simulations. There is a clear positive correlation within the blue data showing higher galaxy mass equals a higher black hole mass.

2.0.4 Super-Eddington Accretion

One important way BH grow is via Eddington accretion. To understand super-Eddington accretion we must first discuss the Eddington limit. This defines the theoretical limit to black hole growth based on the maximum luminosity a black hole can maintain while balancing gravitational pull and radiation pressure ([Marziani et al., 2025](#)). Expressed as Eddington luminosity, the equation for the Eddington limit looks like this

$$L_{Edd} = 4\pi GMc/\kappa \quad (2.1)$$

where G is the gravitational constant, M is the mass of the black hole, c is the speed of light, and κ depends on the opacity of the accreted material. The Eddington Limit can be surpassed in two ways, via non-spherical accretion, as seen in a disk, or when photons are trapped and advected inward with the gas, reducing radiation pressure. Circumventing the Eddington limit would

be required to create a [supermassive black hole \(SMBH\)](#) via only accretion. A standard stellar mass black hole might range from a few Solar masses to a few hundred Solar masses but that is nothing compared to the SMBHs that reside in the million Solar mass range.

2.1 AGN Structure & Categorization

2.1.1 AGN Categorization

AGN can be generally understood by a unified model. Many of the separations between categories of AGN are arbitrary as all AGN are fundamentally comprised of the same components: a supermassive black hole with an accretion disk ([Urry, 2004](#)). This disk can radiate light from the optical to the X-ray. The gas in the accretion disk and around the black hole is essential to the unified model; lower velocity gas lies in the narrow line region and higher velocity gas in the broad-line region. The broad-line region is close to the central region of the AGN believed to be located on the inside of the dusty torus while the narrow-line region is located farther out, a few pc. The gas and dust around an AGN also usually form a obscuring torus that can hide the broad-line region from some observation angles. Finally, it is common for AGN to have relativistic jets that form within 100 Schwarzschild radii of the black hole that can extend out from the black hole for tens of kiloparsecs or farther into the megaparsec range. Figure [2.5](#) depicts these properties.

AGN are generally categorized into two groups: Seyfert AGN and quasars. One of the main distinctions between the two is distance. Seyfert galaxies are generally nearby while quasars, and their subcategory of blazars, are very distant objects. We are looking for quasars with a combination of the [James Webb Space Telescope \(JWST\)](#) and the [Very Large Array \(VLA\)](#), and Chandra data. Quasars are highly energetic AGN actively undergoing feedback processes that produce large relativistic jets that can be easily detected by telescopes. Blazars are a sub-class of quasars that have their jets in line with the observing telescope, so the jet is recorded head-on. Within the Seyfert categorization there are two main subgroups, Type I and Type II. Both narrow line and broad-line emission lines are visible on spectra of Type I Seyfert AGN. This differs from Type II where the BLR is obscured. Figure [2.4](#) contains a breakdown of AGN structure showing the scale of these, and other, AGN features.

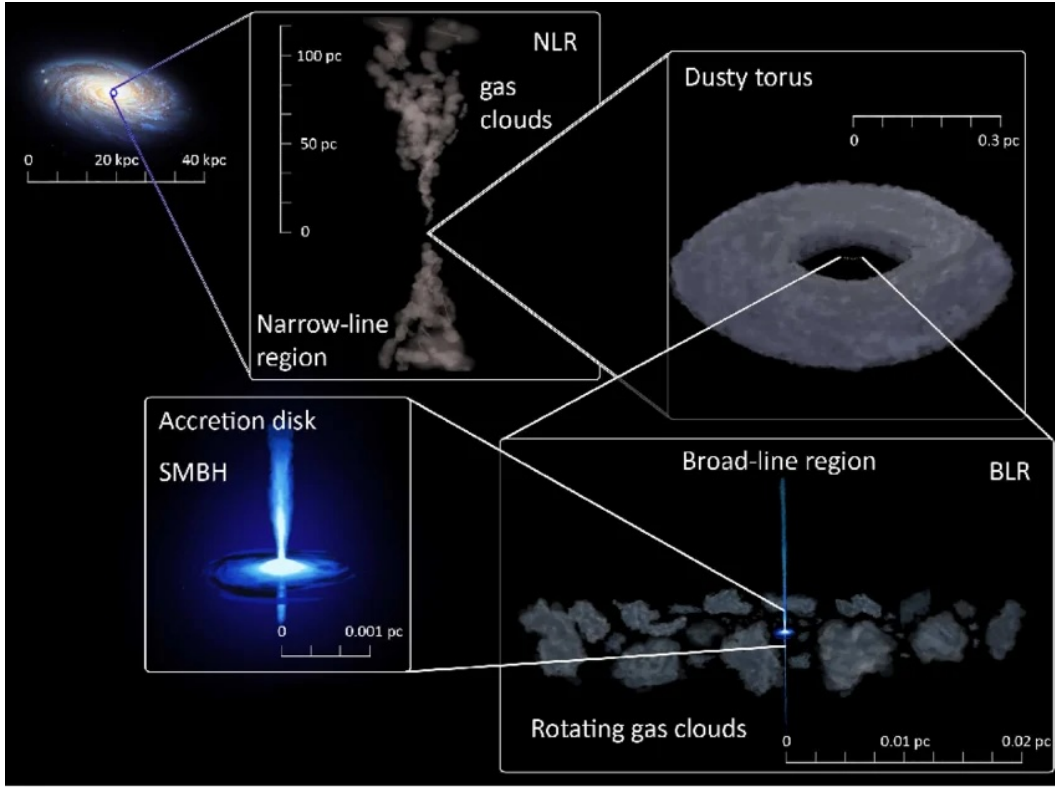


Figure 2.4 A breakdown of AGN structure, highlighting the NLR, dusty torus, BLR, and accretion disk with relative sizes (Malygin et al., 2020). In the first frame we can see how small an AGN actually is compared to the entire galaxy.

Radio galaxies are another type of active galaxy that possess gas jets extending far from the central nucleus. These jets emit synchrotron emission that we detect in radio wavelengths. Synchrotron emission occurs when electrons are accelerated by a magnetic field.

The main differentiation between these categories is the orientation of the AGN as in Figure 2.5. Further more precise classifications can be defined by: composition and densities of the surrounding gas and dust; the rate of accretion; the presence of various AGN feedback processes; and spectral features.

2.1.2 Structure

A dormant supermassive black hole lurks in the center of most galaxies, but when infalling matter feeds the black hole it bursts to life, emitting bright

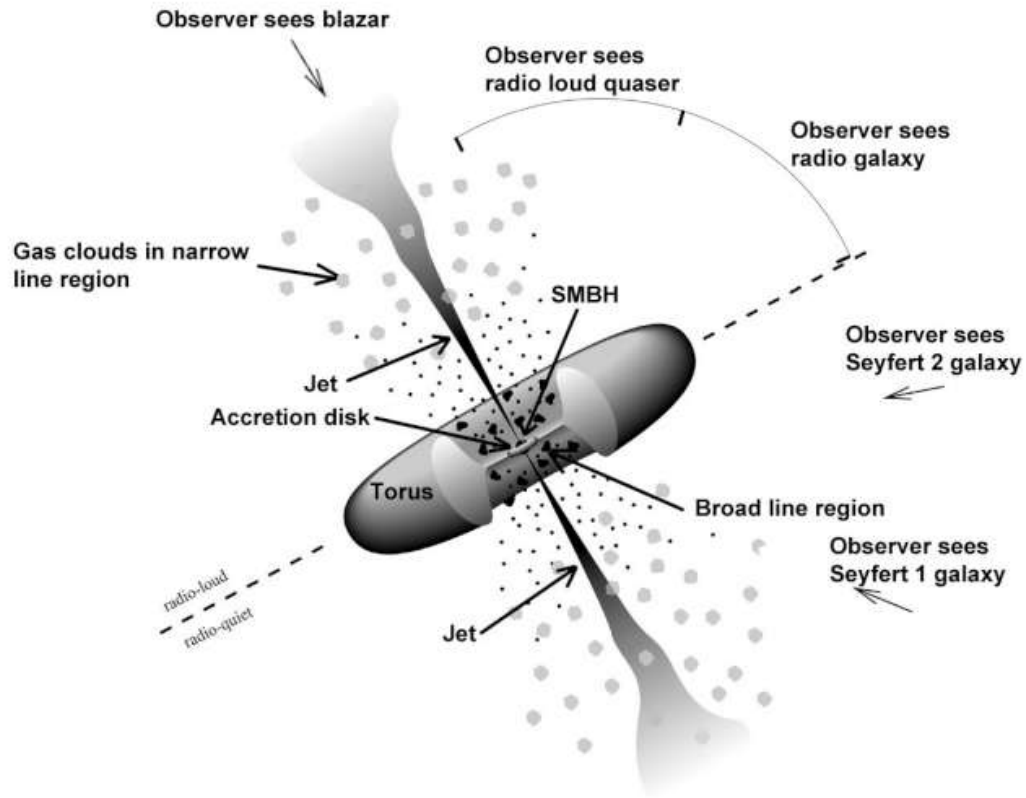


Figure 2.5 A diagram showcasing the unified model of AGN and how the viewing angle easily influences its categorization. While not every AGN has these characteristics they are commonly found with most AGN. *Image Credit: NASA*

flashes of X-rays and radio. A SMBH that has a consistent and constant source of infalling matter is considered an active galactic nuclei (Shields, 1999). The source of all interactions between an AGN and its host galaxy stem from the infalling matter, which forms a rotating disk around the AGN called an accretion disk. These disks are made of hot luminous gas and dust that lose gravitational potential energy as they spiral inward, eventually falling into the BH forever (Shields, 1999). There are two main models of accretion disks: a “hot” and “cold” model. The former is characterized by a low accretion rate and a thick, radiatively inefficient puffy disk. The latter by a thin radiatively efficient disk with a high accretion rate (Hlavacek-Larrondo et al., 2022). Regardless of the type of disk, an immense amount of energy is released from the disks as the AGN consumes the surrounding gas and dust.

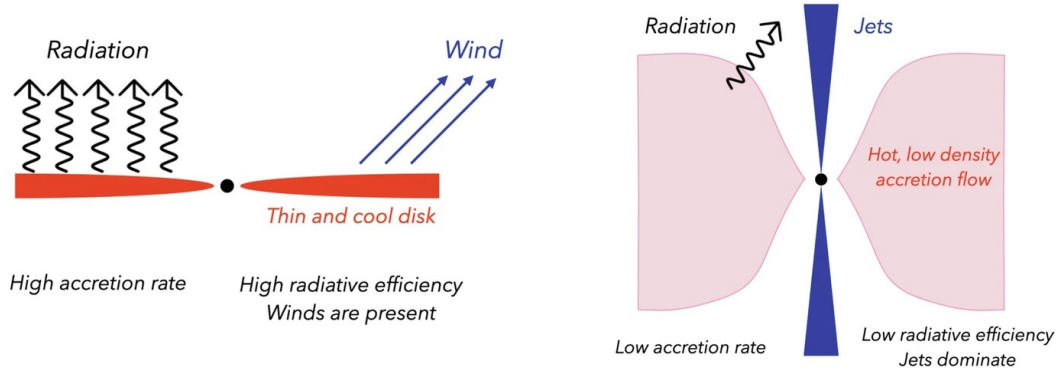


Figure 2.6 Figure 5 from [Hlavacek-Larrondo et al. \(2022\)](#) depicting high and low accretion rates. High accretion rates create a thin disk that produces large amounts of radiation and fast, large angle winds. Low accretion rates create a puffy disk with minimal radiation and relativistic jets.

2.1.3 AGN Feedback

The released energy from an accretion disk is called AGN feedback. Feedback occurs in two main modes: the first is radiative or quasar mode, and the second is kinetic or radio mode. In quasar mode, feedback is dominated by radiation and winds. Conversely, radio mode is dominated by radio jets and x-ray cavities ([McNamara & Nulsen, 2012](#)). It is very common for AGN to exist in some combination of feedback modes, which is why there are so many multi-wavelength tracers for AGN, and why we expect to find AGN with a combination of infrared and Radio detections.

Radio Mode

Radio mode is also referred to as kinetic mode. The jets that power the activity in this mode are relativistic radio bright jets of plasma. Both the x-ray cavities and the shocks originate from these jets. X-ray cavities (also referred to as bubbles) appear as dark spots on opposite sides of an AGN, as seen below in [Figure 2.7](#).

X-ray cavities are formed around jets as radio bright heated gas (plasma) is pushed outward by the jets. At first the bubble expands adiabatically, creating the weak shocks in the surrounding gas seen in [Figure 2.8](#). The expansion of the bubble slows, but doesn't stop, as the size increases. Eventually the pressure of the ICM matches that of the bubble and at this point, the energy provided by the AGN to inflate the bubble is equal to the enthalpy of the bub-

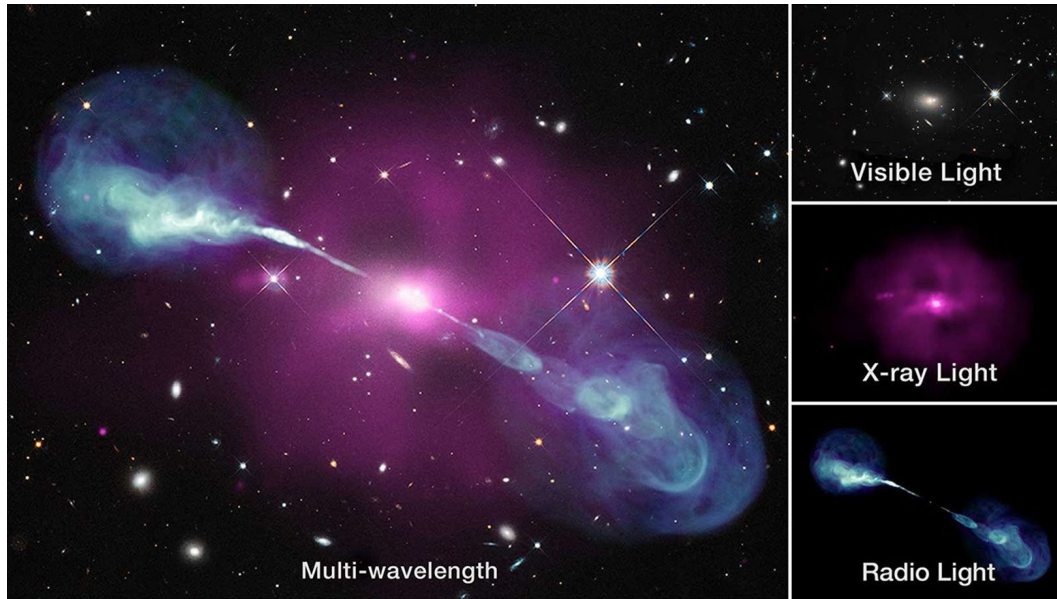


Figure 2.7 Hercules A, a radio bright galaxy, is shown in multiple wavelengths: Visible light revealing a elliptical galaxy. X-rays show a cloud of gas heated by the AGN along with two bubble shaped cavities. Radio reveals striking jets spanning almost 1 million light-years in length. The jets have shifted since the formation of the radio bubbles causing a misalignment ([Blandford et al., 2019](#)). *Credit: X-ray: NASA/CXC/SAO; visual: NASA/STScI; radio: NSF/NRAO/VLA.*

ble ([Hlavacek-Larrondo et al., 2022](#)). The effects of buoyancy take over when the hot plasma is underdense compared to the ICM. The bubble rises through the ICM, away from the AGN, staying in approximate pressure equilibrium with the surrounding gas, as its volume gradually increases. This effectively converts the bubble's thermal energy into potential energy of the displaced gas ([Hlavacek-Larrondo et al., 2022](#)). This ensures all bubble energy will be released to the ambient gas.

Quasar Mode

Quasar mode is also referred to as radiative mode. Feedback in this mode is powered by large amounts of radiation from the accretion disk. This powerful radiation comes from the energy lost by infalling material as it spirals into the AGN. The thin cool disks seen in quasar mode are more conducive to fast, wide angle winds than radio mode. Powerful radiation released from the infalling

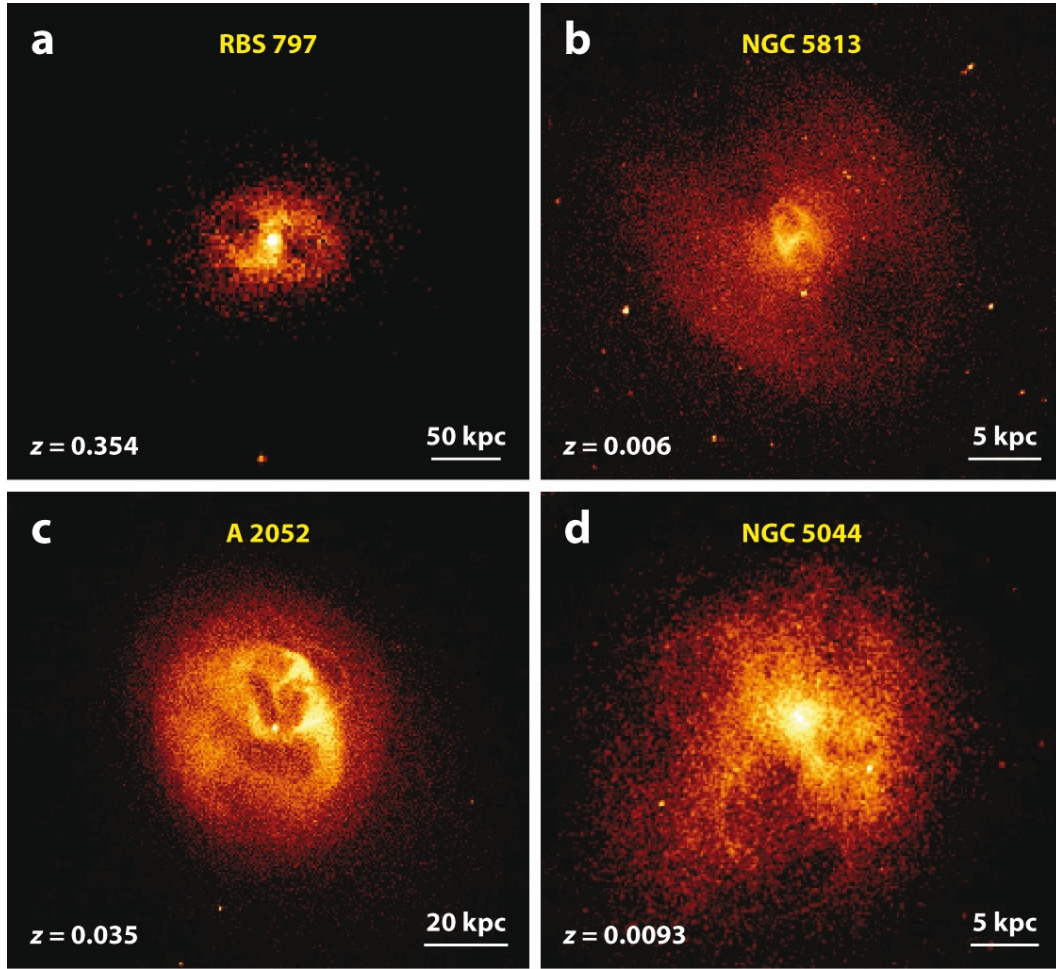


Figure 2.8 Chandra X-ray data showing AGN-gas interactions over various scales (Fabian, 2012). Note the differing scales between the four examples. Image a and b both have clearly defined x-ray cavities, b specifically has two at different stages of expansion.

material also contributes to the acceleration of these winds. Wind speeds above 10,000 km/s have been found with spectral analysis of absorption lines from AGN (Hlavacek-Larrondo et al., 2022).

Outflows from AGN can be seen in both radio and quasar mode, although they behave differently. In radio mode, they are small, traveling alongside jets, usually not gaining enough speed to reach escape velocity of the host galaxy (Hlavacek-Larrondo et al., 2022).

In quasar mode, outflows are highly multiphase and include: neutral atomic gas outflows with velocities reaching up to 1500 km/s; mildly or partially ionized gas, so called “warm absorbers,” with velocities in the few hundreds of km/s; ionized outflows reaching several thousand km/s; and even highly ionized outflows reaching relativistic velocities of 0.2-0.4c. Outflows in this mode often reach out to kiloparsec scales, allowing them to have an effect across the entire galaxy (Hlavacek-Larrondo et al., 2022).

2.1.4 Star Formation & Quenching

AGN feedback processes can heat the surrounding gas to millions of Kelvin and these temperatures can be high enough to prevent or quench star formation, which occurs at around 20 Kelvin. The radiative pressure of AGN feedback on the dust and gas within the galaxy creates outflows that sweep the gas and dust in the galaxy into outflowing shells. This process can expel the cold gas reservoirs from the galaxy, as we see in 2.9, greatly reducing or quenching star formation (Ishibashi et al., 2018).

2.1.5 The Unified Model of AGN

Today’s unification schema of AGN has advanced since the 1980s in several important ways. Netzer (2015) summarizes these developments:

1. They concluded that most AGN tori were probably axisymmetric and clumpy although the models of the time were somewhat unsatisfactory, with hard to constrain parameters. The simulations of the time were more limited to AGNs with small black holes and did not include all the processes involved.
2. Models of disk wind had expanded to explain more properties of the obscuring processes (eg. the formation and distribution of dust clumps

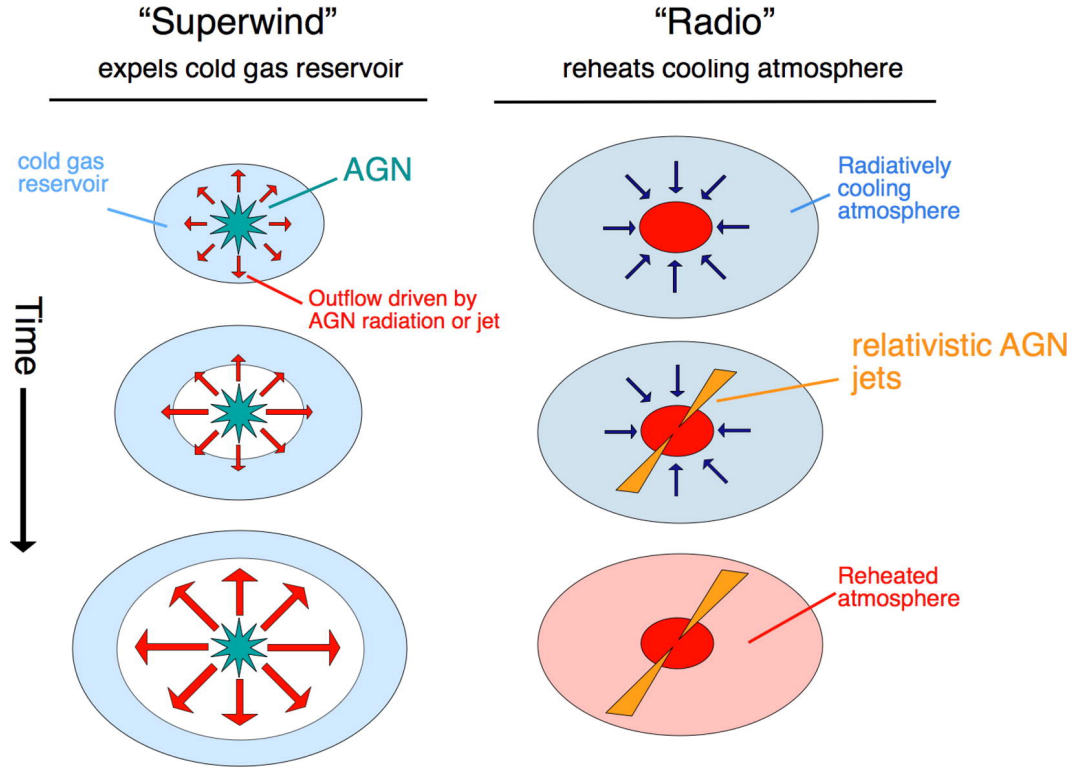


Figure 2.9 A schematic that describes two main modes of AGN feedback we’ve discussed and how they can influence their surroundings over time. ”Superwind” mode is another name for quasar mode. Figure 1.8 from [Harrison \(2016\)](#) and Figure 7 from [Alexander & Hickox \(2012\)](#).

where expected). All models struggled to form and maintain column-density clumps with no clear preference for radiation-pressure driven or magnetic driven winds.

3. The structure of AGN tori was still under debate because the covering factor and anisotropy of the torus were not fully understood. [Netzer \(2015\)](#) discusses the disagreements between the methods of describing this relationship. The four methods they mention involve decreasing f_C with increasing L_{AGN} , the L_{dust}/L_{AGN} ratio, on EWK(K_α), or splitting the tori into groups based on (the presence of?) X-ray-absorbing columns.
4. New observations were able to provide more insight into the structure of the torus and the properties of the NLR gas, specifically bicone observa-

tions and IR interferometry. These observations allowed for a link to be made between NLR gas kinematics and defining torus boundaries. IR interferometry encouraged a reconsideration of the role of MIR-emitting-polar outflows.

5. The widely varying classifications of AGN also complicate the idea of a unified model. Many subcategories will not necessarily fit into a unified model and [Netzer \(2015\)](#) mentions some subgroups that may not belong within the unified model at all. Among them are type-II AGN, which confuse many issues, including the fraction of obscured and unobscured low luminosity AGN ([Netzer, 2015](#)).
6. The final major note on unification model developments is that it doesn't fully account for BH evolution. Specifically, the behavior of merging systems that likely vary greatly from that of an isolated host complicate the model.

2.2 Previous Measurements of Point-Source Features within Galaxies with JWST in the North Ecliptic Time Domain Field

In this study we are expanding upon the work of [Ortiz et al. \(2024\)](#) by increasing the sample of galaxies examined through morphological fitting. The authors suggested that AGN can be identified by the presence of a [point spread function \(PSF\)](#) within a galaxy ([Ortiz et al., 2024](#)). Their selected sample size of 66 galaxies (referred to as Central Point-Source Galaxies or CPGs) produced 16 with measurable point-source components (specified at PS-CPGs, Point-Source Central Point-Source Galaxies). In this paper we adopt similar terminology to differentiate the classifications of our sample of galaxies. To expand on their conclusions, we gathered a collection of 908 galaxies from the data collected in the COSMOS-Web field ([Shuntov et al., 2025](#)).

[Ortiz et al. \(2024\)](#) identified 14 of the 16 PS-CPGs with `GalFit` that were host to IR-luminous AGN. The 14 PS-CPGs were classified as host to IR-luminous AGN based on their $m_{F365W} - m_{F444W}$ color. They also found that `GalFit` ([Peng et al., 2002, 2010](#)) preferred a Sérsic profile with a PSF component fit over that of a double-Sérsic fit. The authors determined this preference by comparing χ^2_ν values for both morphological fits. `CIGALE` ([Boquien et al.,](#)

2019) found a $\langle f_{AGN} \rangle = 0.47$ which was higher than their control sample. They confirmed that all 14 PS-CPGs have an unresolved PSF in the images that motivated the work, providing evidence that the presence of an unresolved PSF within a galaxy in the near-infrared (NIR) supports the presence of an AGN.

Chapter 3

Methods

We began our research into dust-obscured AGN by selecting data from JWST’s [Near-Infrared Camera \(NIRCam\)](#) and VLA, as well as Chandra detection flags from the COSMOS-Web 2025 DR1 ([Shuntov et al., 2025](#)). That calculated photometric redshifts for all of our targets and magnitudes in all NIRCam filters. This would allow us to study the morphological characteristics of dust-obscured AGNs through color plot and analyze the redshift distribution of our sample. We devised python code using `pysersic` to fit the AGN to two templates: a Sérsic profile with a point-source component and a double-Sérsic profile. Each CPG was analyzed in $5'' \times 5''$ cutout around the center of the detection. For some radio detections this meant our data were unusable, which we discuss in [3.1.2](#).

3.1 Data Selection

We use data from the James Webb Space Telescope and its Near-Infrared Camera and the Very Large Array for the majority of the analysis in this work. We also use Chandra detection flags from the COSMOS-Web 2025 DR1 ([Shuntov et al., 2025](#)).

3.1.1 JWST/NIRCam

The reduced NIRCam mosaics and photometric data we use in this work come from the COSMOS-Web 2025 Public Data Release 1 ([Franco et al., 2025](#);

Shuntov et al., 2025). We are focusing on NIRCam mosaics from detector B4 with a 30 milliarcsecond pixelscale in the F444W band. All observed targets lie within the field of the NIRCam B4 mosaic, we trimmed down all data from other sources to match the [field of view \(FOV\)](#) of detector B4. The B4 mosaic covers a section of the sky $576.36'' \times 747.36''$.

Visually Identified Sources

We identified the original subset of targets within the selected NIRCam field. We identified targets by eye that had an unresolved PSF consistent with a JWST PSF. These targets make up just over 40% of the targets in our sample. The other 60% we identified via radio detection, as discussed in section 3.1.2. Our visual identification process involved splitting the mosaic into smaller sections of about $40'' \times 20''$ and scanning each section top to bottom, left to right with a ruler to guide. Any PSFs identified along the way were selected by drawn region.

Target data from COSMOS-Web DR1

Along with the mosaics used in this project, the COSMOS-Web team put together a master catalog of Photometry (PSF-homogenized Aperture and SE++ Model-Based), LePhare Photometric Redshifts and Physical Parameters, Non-PSF-homogenized Aperture Photometry, CIGALE Physical Parameters and Non-Parametric SFH, Machine Learning Morphological Classification, and Bulge+Disk Morphological Measurements & Photometry (Shuntov et al., 2025). This catalog has a total of 784,016 sources with various flags to describe the quality of the sources.

The morphological measurements done by the COSMOS-Web team contain Sérsic & Bulge + Disk morphological fitting but does not contain the Sérsic + PSF morphological fitting done by `pysersic` in this work.

3.1.2 VLA

We identified the radio sources with the help of data from The CHILES Continuum and Polarization (CCP) Survey, an ultradeep continuum imaging study of the COSMOS field conducted using the Karl G. Jansky Very Large Array (Gim et al., 2025; Luber et al., 2025). This survey utilizes a synthesized beam of $5''.5 \times 5''.0$. This work uses the 523 radio detections from the CHILES 1.4 GHz radio continuum source catalog of 1678 sources that are within the NIR-

Cam B4 mosaic. Combining the radio and the visual detections left us with 908 targets to further investigate. We used a python program to extract the region centers from the region file. We created a small $5'' \times 5''$ cutout around the center of each target's region file. Due to the resolution of the survey beam, approximately 18% of radio detections, those that were not centered on a target were lost when performing the morphological fitting. Many of these off center targets are due to the beams resolution reading multiple NIRCcam sources as one VLA source, as seen in Figure 3.1. In cases such as this, the program does not perform a proper fit and the fitting fails.

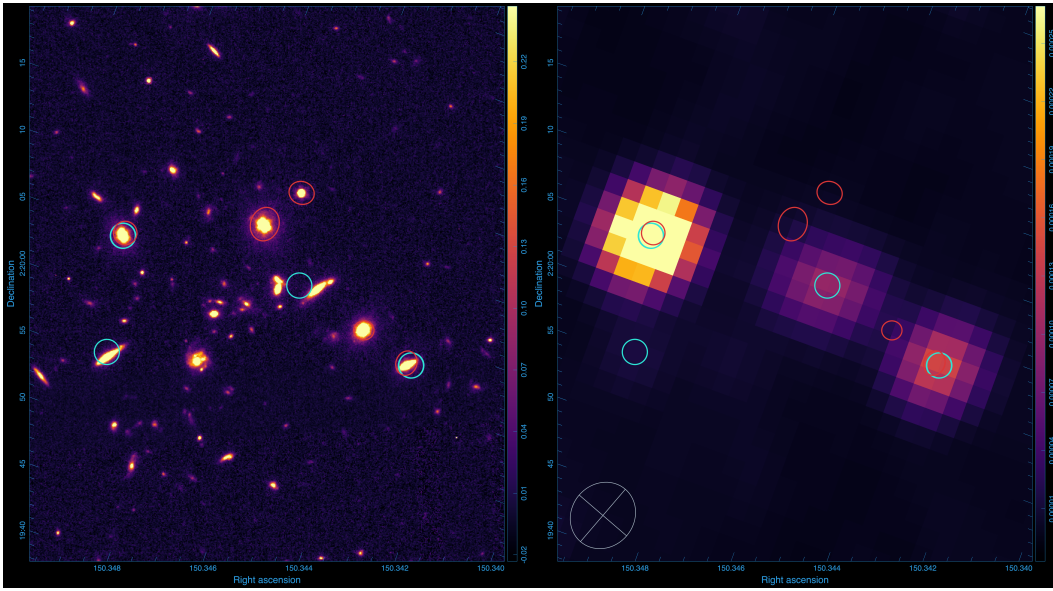


Figure 3.1 A comparison of JWST/NIRCcam data (*left*) and VLA data (*right*). Both images have the same scaling. Regions are overlaid representing visually selected targets in red and radio VLA detections in blue. The resolution of the radio data is worse than that of JWST and you can see the cluster of 50 or so galaxies in JWST are condensed into only three radio detections.

3.2 Morphological Fitting with pysersic

We ran all selected targets through the morphological fitting program, `pysersic` (Pasha & Miller, 2023), to model the light distribution. Following Ortiz et al. (2024) we compared two models, a Sérsic profile plus a PSF component and a double-Sérsic model, to determine if the targets contained a true point source. In many cases, this greatly generalizes the characteristics and components of

these galaxies, but in most cases it satisfactorily characterizes point-like features.

The Sérsic + PSF profile fit performs a standard Sérsic profile fit with an added point-source component within the galactic nucleus of the target. The double-Sérsic profile fits a standard Sérsic profile twice, the first on an inner radius and the second on the outer radii of the target. The inner Sérsic profile of the double-Sérsic fit is generally comparable to the PSF component of the Sérsic + PSF fit.

A Sérsic profile has the flexibility to represent many different galaxy types, making it ideal for the goal of characterizing the selection of NIRC*am* targets. The Sérsic profile is parameterized as

$$I(R) \propto F_{total} \exp \left\{ \left[\left(\frac{R}{R_e} \right)^{1/n} - 1 \right] \right\} \quad (3.1)$$

where F_{total} , the total flux, half-light radius, R_e , and Sérsic index, n , are the parameters of interest to be fit and subsequently used to characterize a galaxy’s morphology (Pasha & Miller, 2023). Based on Ortiz et al. (2024) we assume that the Sérsic profile models the nucleus and extended stellar disk of a galaxy in a double-Sérsic profile fit. For a two component fit like the Sérsic + PSF model the Sérsic profile models the extended part of the galaxy while the PSF component models an unresolved point in its center.

Both fits require some preliminary setup. For the fitting process we selected a few sets of data for each target and set up the priors for each fit. **pyser** requires the image data, the uncertainty, a data mask, and a model PSF for comparison. This model PSF must be smaller than the size of the data. We worked with a PSF only about 0.2” smaller than the data. We discuss the PSF generation further in section 3.3. Additionally, the data mask that **pyser** expects is a boolean mask to remove the unwanted pixels. The data mask given in the COSMOS2025 DR1 masks out the targets with integer values. We need to convert the mask into a boolean array and then flip it to mask the background instead of the targets.

3.2.1 Light Profile Fitting

`pysersic` has three parameters across the two profiles that measure light distribution: flux, f_1 , and f_{ps} . Both profiles measure the total flux of the target. The difference is how the flux of a possible galactic nucleus is measured. The double Sérsic profile fits the first of the two Sérsic profiles to this inner radius measuring the percentage against the total, giving the value for f_1 . The Sérsic profile with a point-source component establishes an inner radius and compares the flux within that to the total flux, giving the value f_{ps} . Both profiles also measure an effective radius, or half-light radius, r_{eff} , that is used to determine the central flux.

Another helpful fit parameter that `pysersic` provides is the Sérsic index, n . For the double-Sérsic fit, we are given n_1 and n_2 , the indices of the inner and outer Sérsic profiles respectively. The value of these indices can give another indicator of the uniformity of the light distribution of the source. An index between 0.65 and 8.00 indicates a generally uniform light distribution (Pasha & Miller, 2023). The higher the value of the Sérsic index is, the more concentrated the light is. Conversely, the lower the value, the less concentrated the light is.

The other parameters of both profile fits refer to the morphological characteristics of each profile: `xc`, `yc`, `theta`, and `ellip`. The only difference between the two profiles is that the double-Sérsic has two Sérsic profiles to fit, inner and outer, labeling the parameters as 1 and 2 respectively. These parameters are mostly self explanatory: `xc` and `yc` label the center of the galaxy in pixels, `theta` measures the position angle in radians, and `ellip` measures the ellipticity of the galaxy.

3.3 PSF Generation

We used Python package `STPSF` (Perrin et al., 2025), formerly `WebbPSF` (Perrin et al., 2012), we were able to compute a simulated PSF $5'' \times 5''$ in the JWST NIRCcam F444W filter to act as a point of comparison for morphological fitting.

Generating a PSF with `stpsf` produces four separate extensions: `OVERSAMP`, `DET_SAMP`, `OVERDIST`, and `DET_DIST` in that order. The first and third extensions (both beginning with "OVER") are oversampled by a value specified when setting up the PSF generation, this defaults to four. The

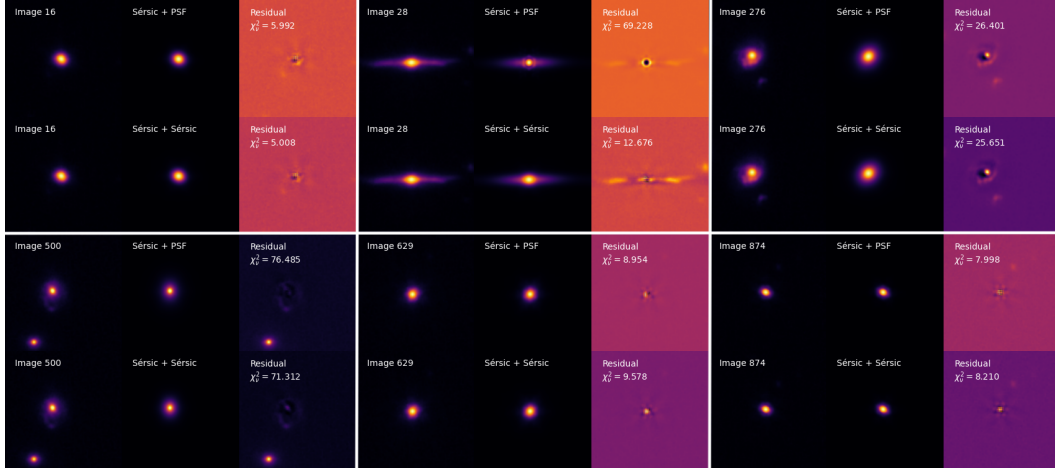


Figure 3.2 An example of a Sérsic + PSF and double-Sérsic fits and their reduced χ^2 values. There are six sub-frames dedicated to each object. From left to right we have the image ID, the fitting method, and the residual with reduced χ^2 . The top panels show the the Sérsic profile fit with a point source component; the bottom panels show the double-Sérsic profile fits.

second and fourth extensions (both beginning with "DET") are not oversampled and use the pixelscale of the specified detector, in this case 30 miliarcseconds. The most important distinction for deciding which extension to use is the difference between the extensions ending in "SAMP" versus the extensions ending in "DIST." The difference here is that the first two extensions, ending in "SAMP," represent the "ideal" or photons-only diffractive effects. The last two extensions, ending in "DIST," have geometric distortion and detector charge transfer effects added to the PSF generation to create a more realistic PSF. Due to oversampling on the third extension, only the fourth extension, "DET_DIST," has both the realistic distortion effects as well as the IPC effect. Given these parameters, the fourth extension is the most fitting for this project as we are comparing this generated PSF to real NIRCcam data that will actually have these detector effects.

3.4 Python Pipeline

We run all raw data from the COSMOS-Web database through a python pipeline of our making. Each target undergoes the fitting process with each profile: a Sérsic + PSF profile fit and a double-Sérsic profile. We use this data to calculate reduced χ^2 values for goodness-of-fit.

We calculate two goodness-of-fit values during the pipeline. The first we calculate from the entire $5'' \times 5''$ frame. The second we calculate by creating a sub-frame (with the same data as the original frame) and analyzing the fit that lies within it. This method excels at weeding out the targets that are misaligned from their radio source in a drastic manner, especially if more than one NIR detections within the frame are emitting in the radio. This can be seen in Figure 3.3 where more χ^2_ν values calculated from the fits within the sub-frame improved compared to the χ^2_ν calculated from the whole frame. This justifies using the sub-frame χ^2_ν values over the values calculated from the whole frame. From here on any reduced χ^2 values, unless otherwise specified, will be those calculated from the sub-frame.

If a target failed any of the fitting we discarded it, along with any targets that did not contain an NIR detection within the $5'' \times 5''$ scope of the cutouts. This only occurred when a radio source was extremely offset from the target or whatever caused the radio source did not have an NIR counterpart. In these specific cases, **pysersic** often still attempted to fit the target and ended up fitting either the corner of the target or fitting some background noise that was not actually data. In these cases, the sub-frame χ^2_ν value would be drastically lower than that of the original. In order to isolate these targets, we created a ratio of the sub-frame χ^2_ν values to the full frame χ^2_ν values. Any target with a ratio of less than 0.03, there are 210 of these targets, would be isolated and visually inspected to identify if the target was being fit by the program properly. In the cases in which it wasn't, we flagged the target in the database. We selected the χ^2_ν ratio cutoff by evaluating a histogram of the reduced χ^2 ratios for both the Sérsic + PSF fit and the double-Sérsic profile fits. Figure 3.4 shows the cutoff for χ^2_ν ratios on a histogram.

We calculated the reduced χ^2 ratio with the reduced χ^2 value by dividing the sub-frame value by the normal reduced χ^2 value. A ratio below one would mean an improved reduced χ^2 value and indicate an improved fit with the sub-frame FOV; vice versa, a ratio above one would mean an increased reduced χ^2 and a worse fit after zooming in, an example of which is in Figure 3.5. This zooming is, notably, not an actual change to the fit but rather just a selection of the middle 1/3 of the data, model, and residual that is then used to recalculate the χ^2 statistic.

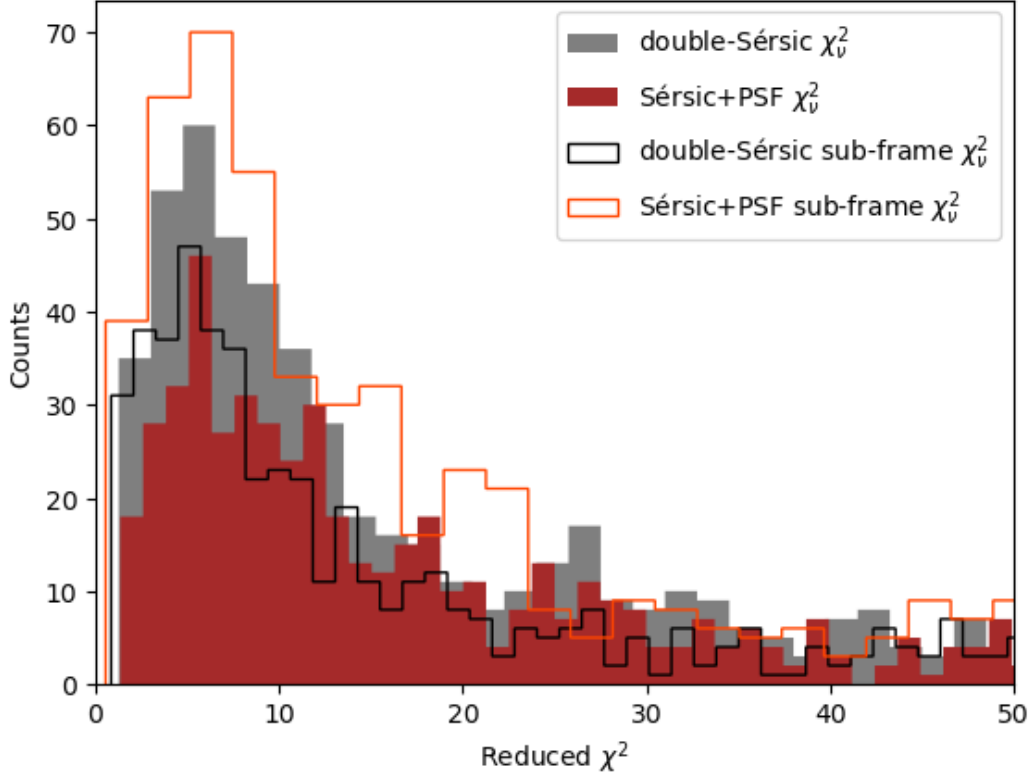


Figure 3.3 Histogram comparing the distribution of the reduced χ^2 values achieved when recalculating reduced χ^2 from a sub-frame to those calculated from the full frame. The filled in histograms show the initial reduced χ^2 distribution from the full-frame images in gray for double-Sérsic and red for Sérsic + PSF. The outlines show the distribution of the reduced χ^2 recalculated from the sub-frame: black for double-Sérsic and orange for the Sérsic + PSF. The distribution of χ^2_{ν} values for the Sérsic + PSF sub-frame improved more than the distribution of χ^2_{ν} values for the double-Sérsic sub-frame calculation.

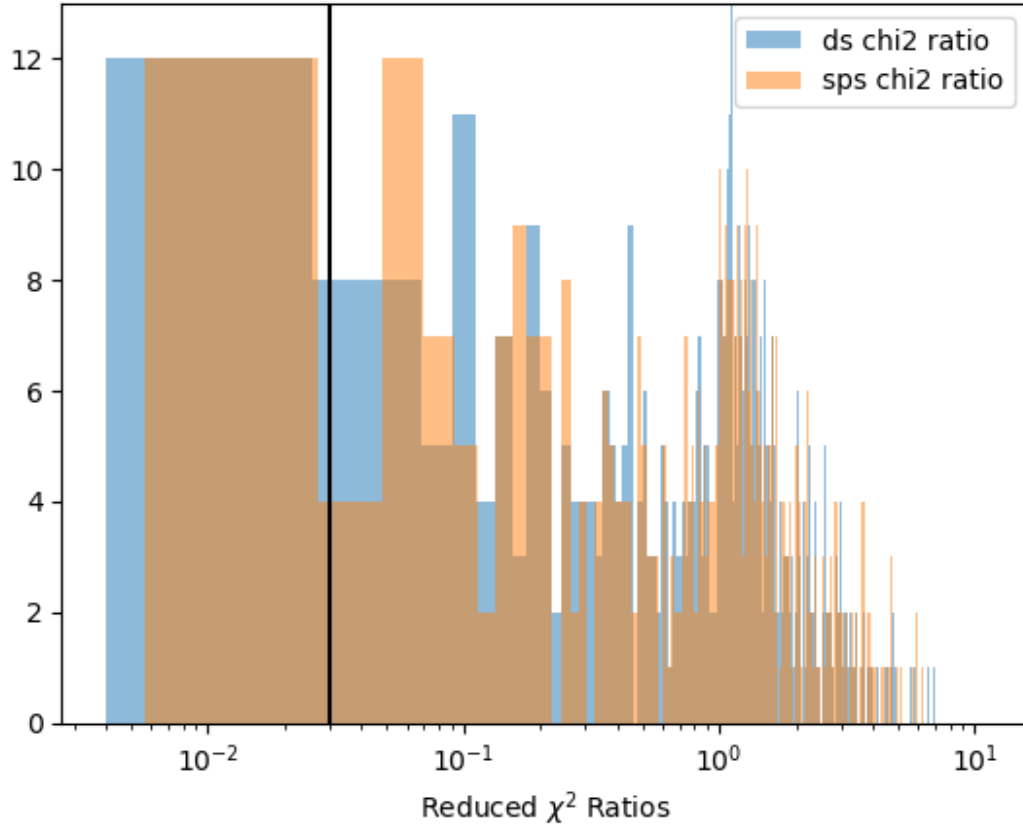


Figure 3.4 A histogram plotting reduced χ^2 values from the two fitting methods: double-Sérsic (ds) and Sérsic+PSF (sps). The vertical line represents the cutoff point, 0.03. χ^2 ratios below 0.03 are flagged in the results as possible mismatches. Targets where the sub-frame did not include the intended target and the reduced χ^2 value dropped significantly as a result.

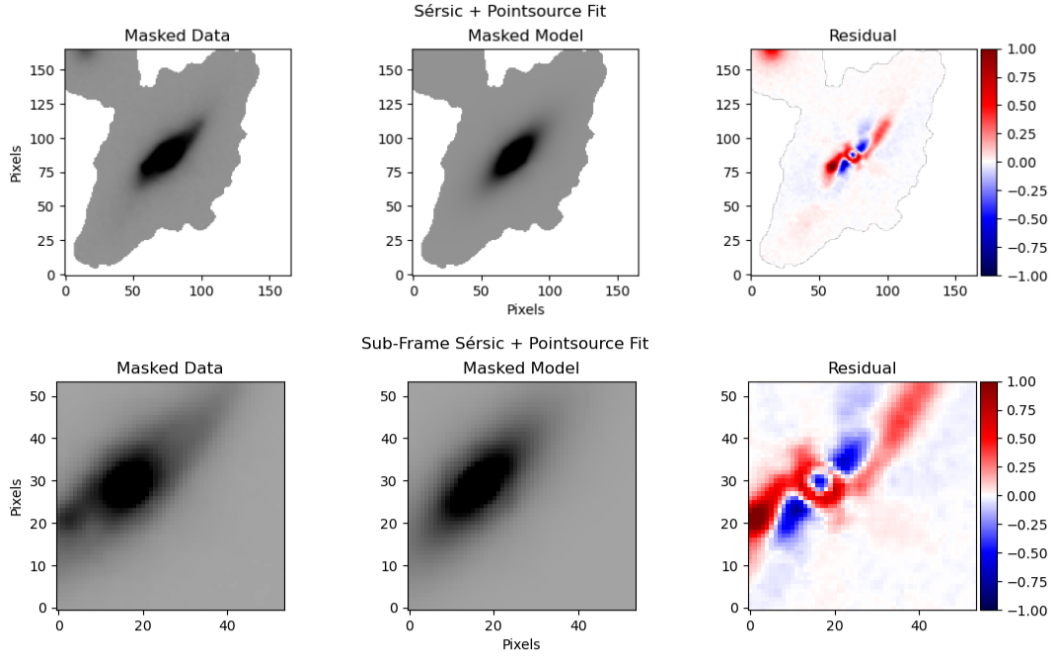


Figure 3.5 Target 238 fit by a Sérsic + PSF profile fit shown in full and sub-frame. The sub-frame does not recalculate the fit, but instead narrows the FOV of the viewing frame to isolate the target from background objects. The reduced χ^2 of the full frame is 25.22 while the reduced χ^2 of the sub-frame is 179.54.

Part II

Project Outcomes

Chapter 4

Results

We identified AGN using our JWST data in two different ways. The first—by identifying unresolved point sources—we discuss in section 3.3. The second—by using multi-wavelength tracers—we discuss in section 4.2.

4.1 Identifying AGN with Measurable Point Sources

After we collected the 727 galaxies with measurable point-sources, we further divided them into PS-CPGs and Bulge-CPGs. Although `pysersic` does not provide a classification of the galaxy nor a χ^2 value after fitting, it is possible to create a comparable analysis by calculating and then comparing χ^2 values. Comparing the reduced χ^2 values calculated, 256, or about $\frac{1}{3}$ of the original targets had a better fit (lower reduced χ^2) when fit by the Sérsic + PSF profile. This would generally indicate that these targets contain a point source and would fall into the PS-CPGs category as defined by [Ortiz et al. \(2024\)](#) to be a point-source central-point-source galaxy. It is reasonable to posit that the majority of sources selected could very well be galaxies with a compact stellar bludge or nuclear starburst, referred to as Bulge-CPGs by [Ortiz et al. \(2024\)](#).

Among these PS-CPGs there are a total of 31 targets with a flux percentage from the central point over 40%. These are a lopsided mix of radio selected targets and those selected for an unresolved point source. Just one of these 31 sources is radio selected. The we chose a cutoff at 0.4 because of a gap in

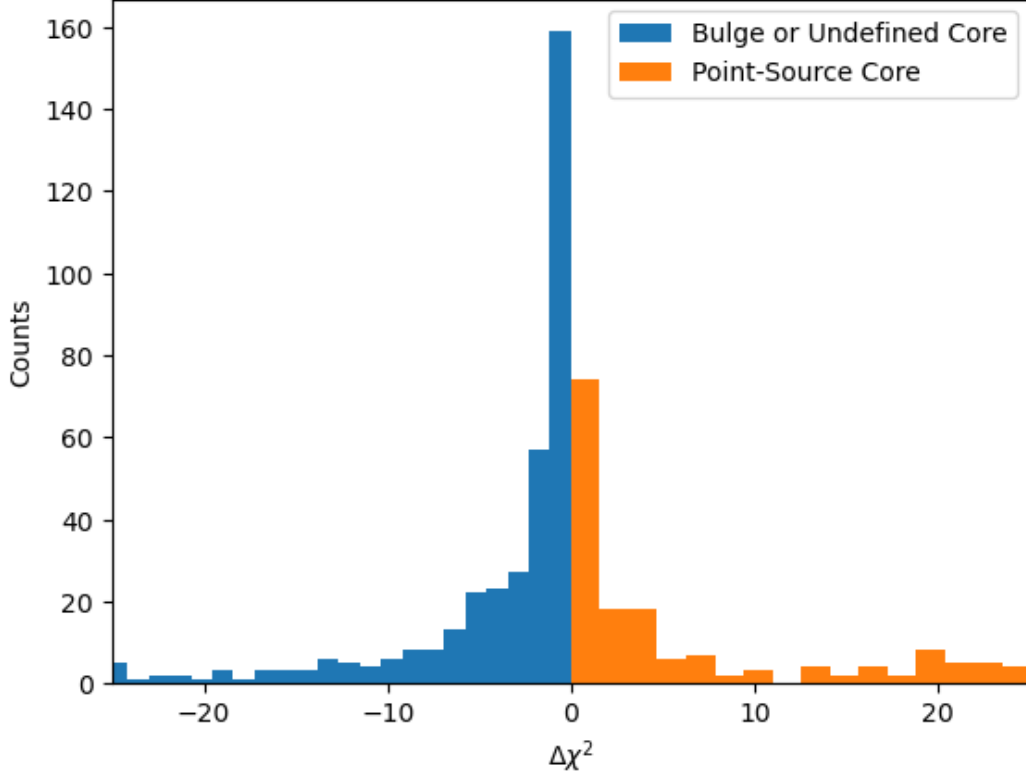


Figure 4.1 Based on Figure 4 of (Ortiz et al., 2024). A Histogram of the χ^2 goodness-of-fit statistic between the double-Sérsic and Sérsic+PSF fits. $\Delta\chi^2 = \chi^2_{\text{double-Sérsic}} - \chi^2_{\text{Sérsic+PSF}}$ indicating a predilection for either a bulge or point-source core.

the histogram of the AGN fraction distribution, as shown in Figure 4.3. There are 197 targets with a f_{AGN} close to 0. Some of these targets appear to have a point-like component, but `pysersic` did not measure a high flux. In other cases the flux from the central point is low or near 0 because there is no target. In this case, discounting the low flux PS-CPGs is a valid method to remove bad data. A majority of these targets lay on or around the 0% flux range. We visually identified 41% (80) in the NIR with an unresolved point source. We identified the other 59% (117) by VLA.

We compared 908 targets that were originally selected with the COSMOS2025 DR1 catalog of detections. Using `astropy.coordinates` we matched our coordinates within 1" to COSMOS-Web sources. We decided to use a 1" separation

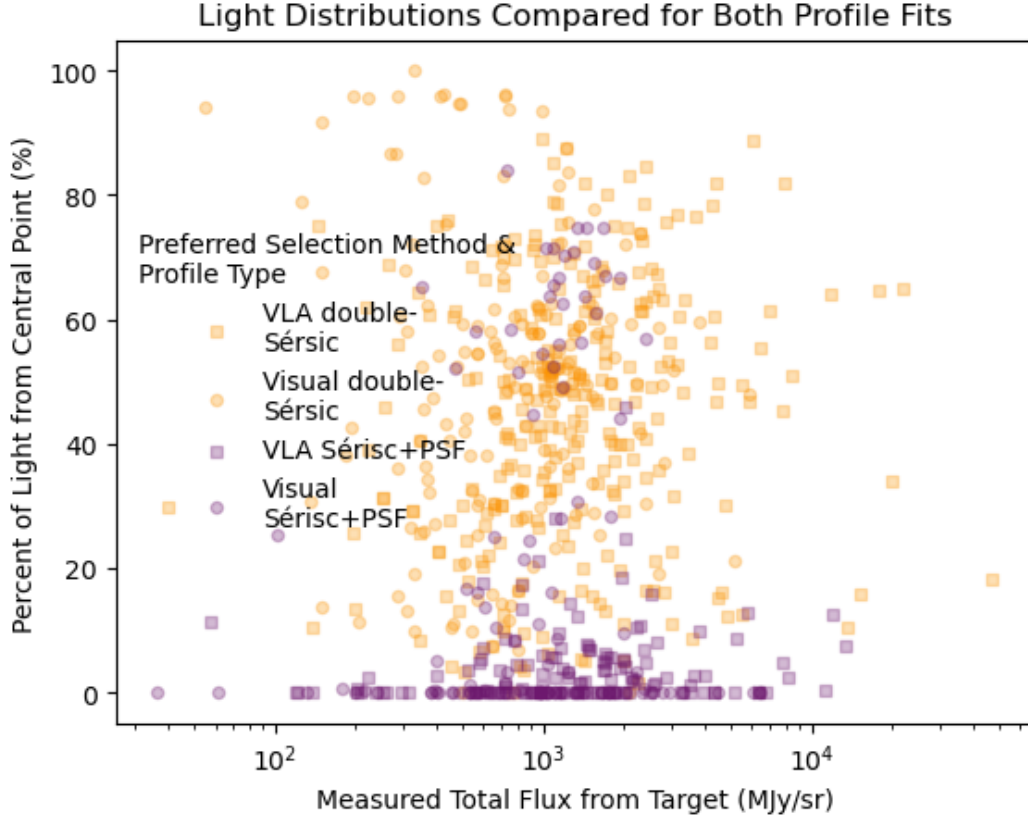


Figure 4.2 This figure compares the total flux measured from the central point vs the total flux measured from the target. The targets that preferred the Sérsic + PSF profile are in purple and those that preferred the double-Sérsic profile are in orange. The VLA identifier shows that the data was radio detected by VLA, whereas the Visual descriptor means the target was identified in the NIRCcam mosaic with an unresolved point source. The purple targets are those with an identifiable point source. Many of these targets lie on or around the 0% range, specifically 197 of the 256 PS-CPGs.

based on the angular size of most galaxies at the median redshift of 1.12 of our targets. The 1'' angular size of the galaxies means that any COSMOS-Web detection that lies within 1'' of our targets are very likely to be the same target, see Figure 4.5. This removed some of the more misaligned or merged radio sources.

All calculated redshifts in the catalog are photometric redshifts, which are

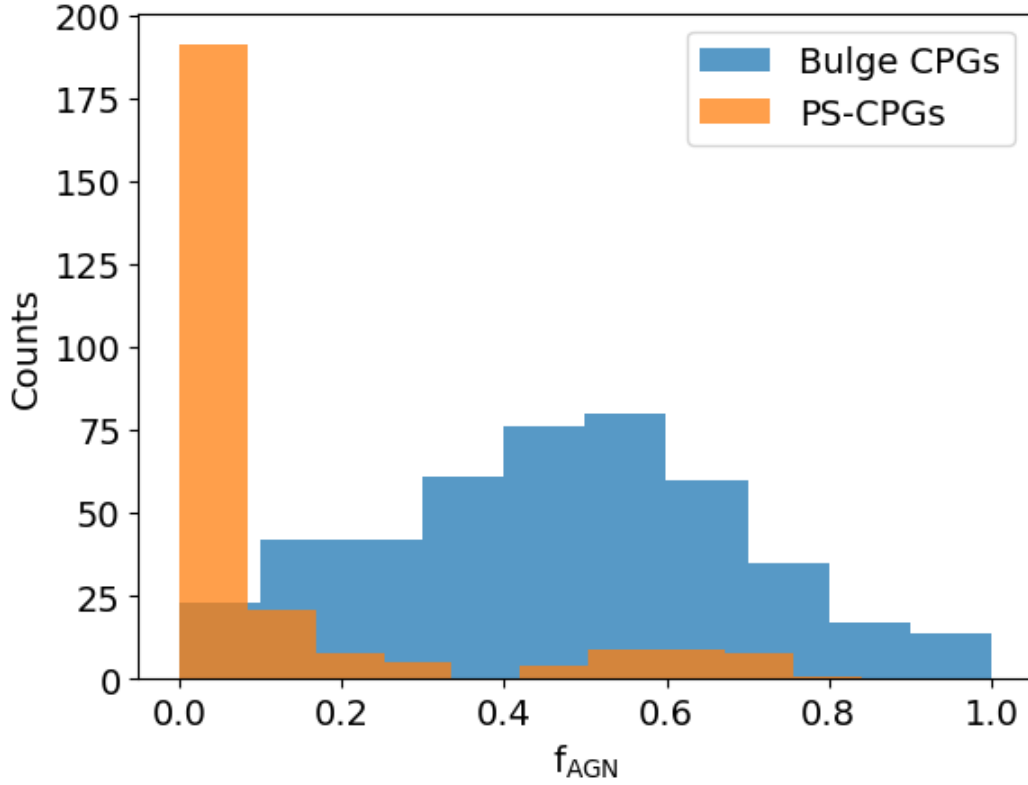


Figure 4.3 A distribution of the AGN fraction of all targets that have an identifiable point-source. There is a gap in the PS-CPG distribution around 0.4 which creates the second population of 31 targets. There are 197 targets with an AGN fraction below 0.1 and 28 targets with an f_{AGN} between 0.1 and 0.4.

only accurate at lower redshifts. We can see in Figure 4.6 that most of the calculated redshifts for our targets (from both models) are below a redshift of about $z = 5$, which is generally trustworthy for a photometric redshift. Those with higher redshifts, especially those at $z = 8$ and above, are definitely untrustworthy. The small clumping in between that range might seem more promising, however, the difference in median redshift isn't significant. The median redshift of all targets below a photometric z of 5 is 1.12 and below a z of 8, the median redshift is only 1.13.

A common AGN selection method is color, and here we compare the COSMOS-Web calculated magnitudes of our targets to identify those that lie above a

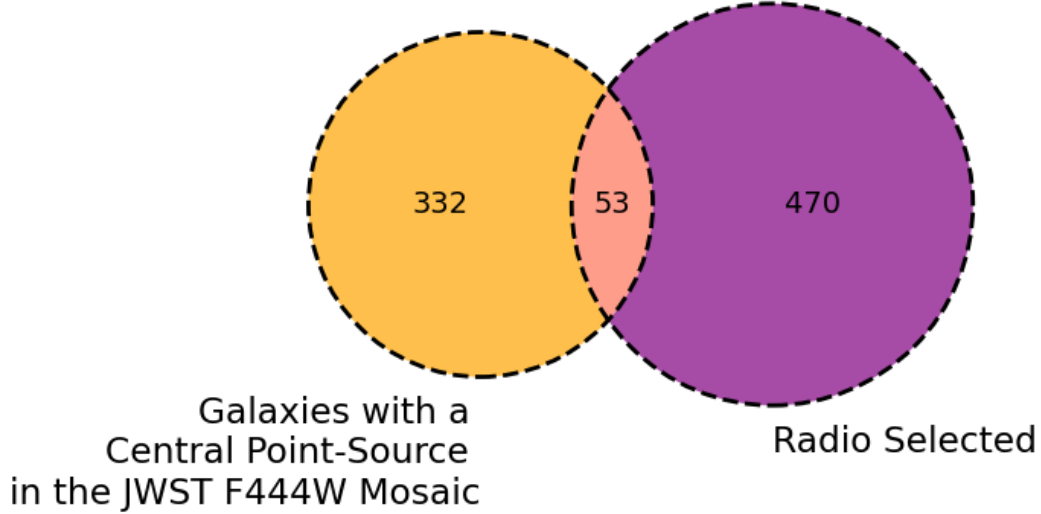


Figure 4.4 This figure depicts how the 908 sources we gathered are split between radio detections, targets with an unresolved point source, and those where the sources overlapped. The sources were considered to be overlapping if they resided within $1''$ of each other.

$m_{F277W} - m_{F444W}$ magnitude cutoff of 0.4, as seen in Figure 4.7. There are 13 PS-CPGs that lie above this cutoff point, and the median f_{ps} of these targets is a very low value of 0.007. Without the color selection, the median f_{AGN} of the 256 PS-CPGs is 0.011, which is a bit higher. These low numbers are a product of the targets that have a very low, near zero, AGN fraction, as seen in Figures 4.2 and 4.3. If the median fractional AGN percentage is calculated ignoring those targets with a value close to zero (anything below 0.1), the median jumps to 0.45, which is a much more reasonable result and much closer to what we expected.

4.2 Multi-Wavelength Tracers of AGN

Our second AGN identification method utilizes NIR, radio, and x-ray detections from JWST/NIRCam, VLA, and Chandra. We retrieved flags for Chandra detections of these targets from the COSMOS-Web 2025 database. This is depicted in Figure 4.8 as a Venn diagram. There are 30 targets that have detections in all three multi-wavelengths tracers that are very likely to be AGN.

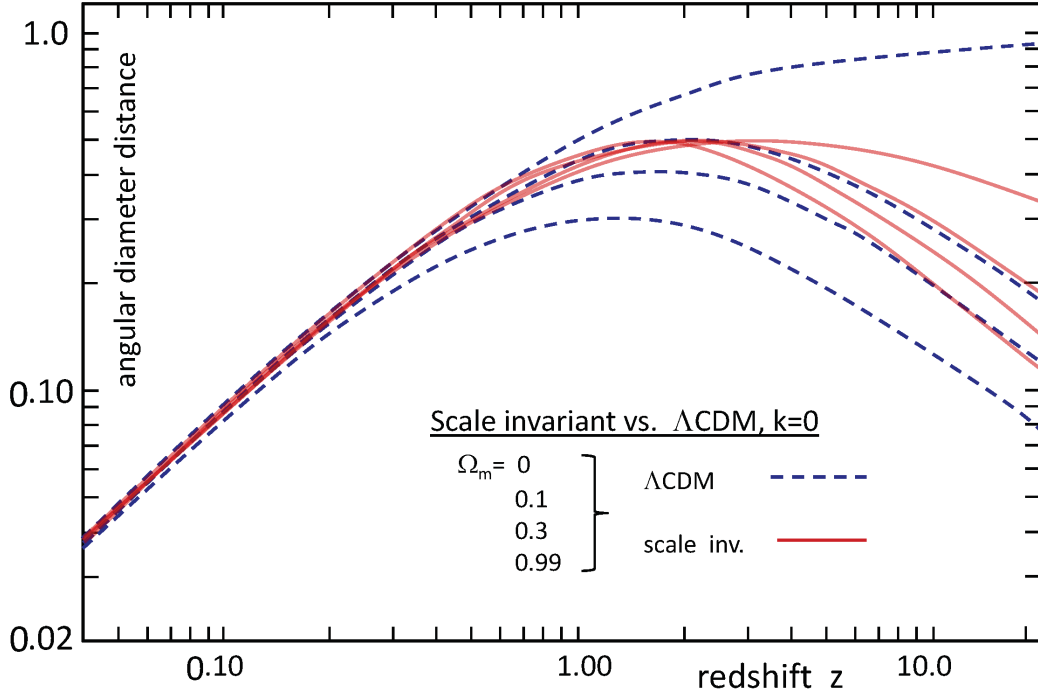


Figure 4.5 Figure 2 from [Gueorguiev & Maeder \(2024\)](#) shows the relationship between the angular size of an object and the redshift for a Λ CDM universe. The angular diameter distance at around a redshift 1 is around $1''$.

Comparing the data of the Venn diagrams in Figures 4.8 & 4.4 reveals that of the 53 targets with both a point source and a radio detection, 13 have a position match in the COSMOS-Web field as well as a Chandra detection. This means that there are 13 targets from our sample likely to be AGN. Because all 13 have matches in the VLA data, as well as having a bright IR central point that has diffraction patterns, we count this as 26 targets out of the initial 908.

With access to the COSMOS-Web database it would be possible to further determine the nature of these targets. There is a flag within the database for confirmed AGN and by this metric, six of the 908 targets are identified as AGN. One of them is included in the 13 position matched targets. Of the six targets identified as AGN in the COSMOS-Web database, only four of the six could be fit, as seen in Figure 4.9. Although two of the four are the same target, as we can see in Table 4.1 by their RA and DEC— one detection in the NIR and one in the radio.

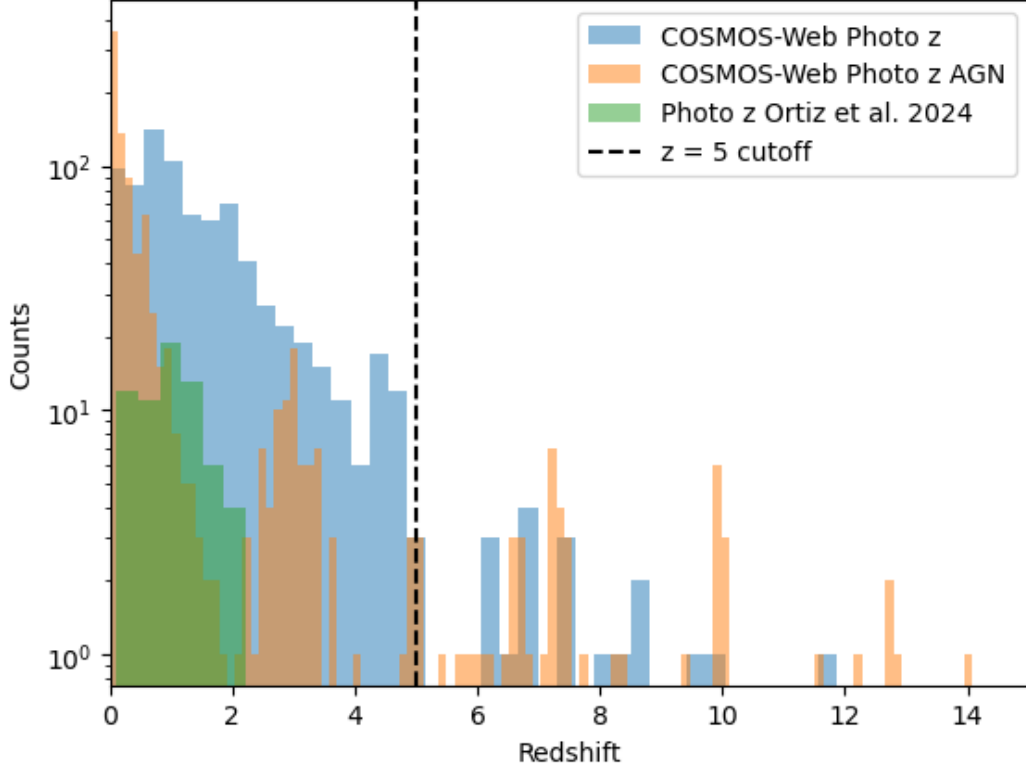


Figure 4.6 The redshifts of all 908 targets were calculated from two models, a standard galaxy model and an AGN model, by the COSMOS-Web team. The standard photo z measurements were calculated from the minimum χ^2 of best fit galaxy template, while the photo z AGN were computed from the best AGN template. (Shuntov et al., 2025)

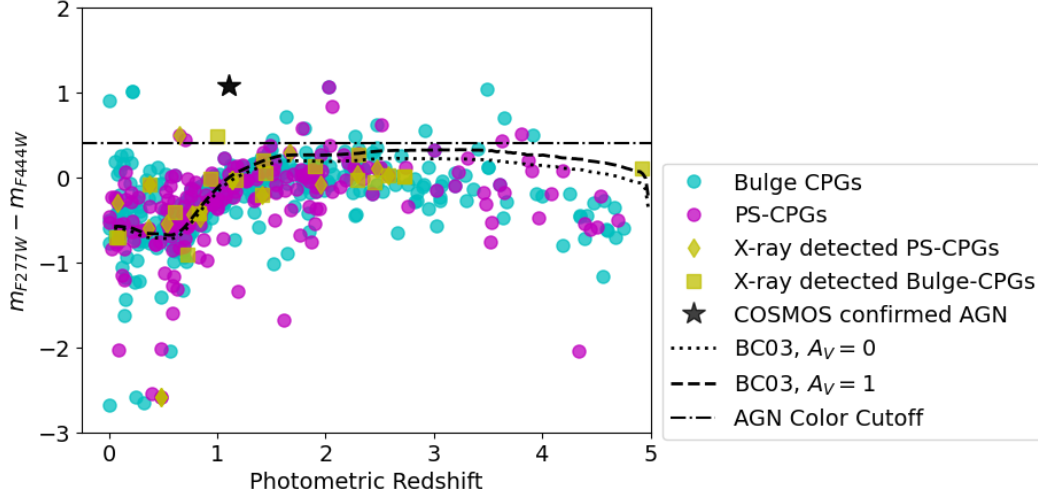


Figure 4.7 A plot of redshift vs $m_{F277W} - m_{F444W}$ magnitude with a AGN color cutoff at 0.4. This plots two models of galaxy color over time with black dashed and dotted lines. One COSMOS-Web confirmed AGN is plotted as a star. A majority of the PS-CPGs identified by **pysersic** fall in line with the rest of the CPG data, only 13 lie above the color cutoff. Five of the 13 fall outside the redshift range of zero to five and as such their colors may be less reliable. A slight majority of X-ray detected CPGs do not show a point-like source in the F444W image, suggesting they are heavily buried and their host galaxy appears as a normal field galaxy.

Table 4.1 Table of four confirmed AGN targets that we have both position matches for and were able to fit with **pysersic**. It is important to note that targets 53 and 739 are the same target, one detected in the Radio and one in the IR. These targets and their corresponding fits are also depicted in Figure 4.9.

Target ID	RA (deg)	DEC (deg)	Photo-z	Photo-z AGN	X-Ray Flag	IR Flag	Radio Flag
53	149.811	2.211	1.103	1.103	1	1	1
211	149.805	2.125	4.864	4.864	0	1	1
336	149.798	2.233	-99.00	0.738	0	1	1
739	149.811	2.211	1.103	1.103	1	1	1

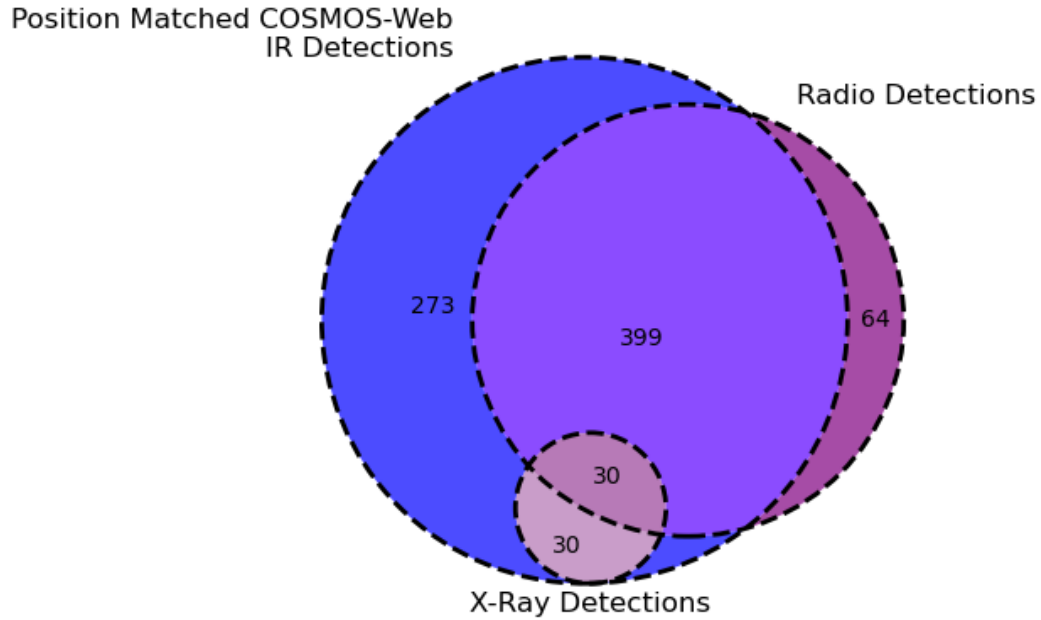


Figure 4.8 A Venn Diagram depicting the the number of sources of the original 908 that were detected in each wavelength. The position matched IR detected sources meant that, of the 908 regions (a combination of VLA data and visually selected targets), 273 had a COSMOS-Web source within 1" of it. The overlap between radio and IR means there are 399 VLA radio sources that had COSMOS-Web detections in the IR. The X-ray detections are from the Chandra flag detections within the COSMOS-Web database. Since all X-ray detections are within the COSMOS-Web database, all X-ray detected targets have IR detections as well. Half of the X-ray detections also have VLA detections. These 30 targets are likely to be AGN based on these three multi-wavelength tracers.

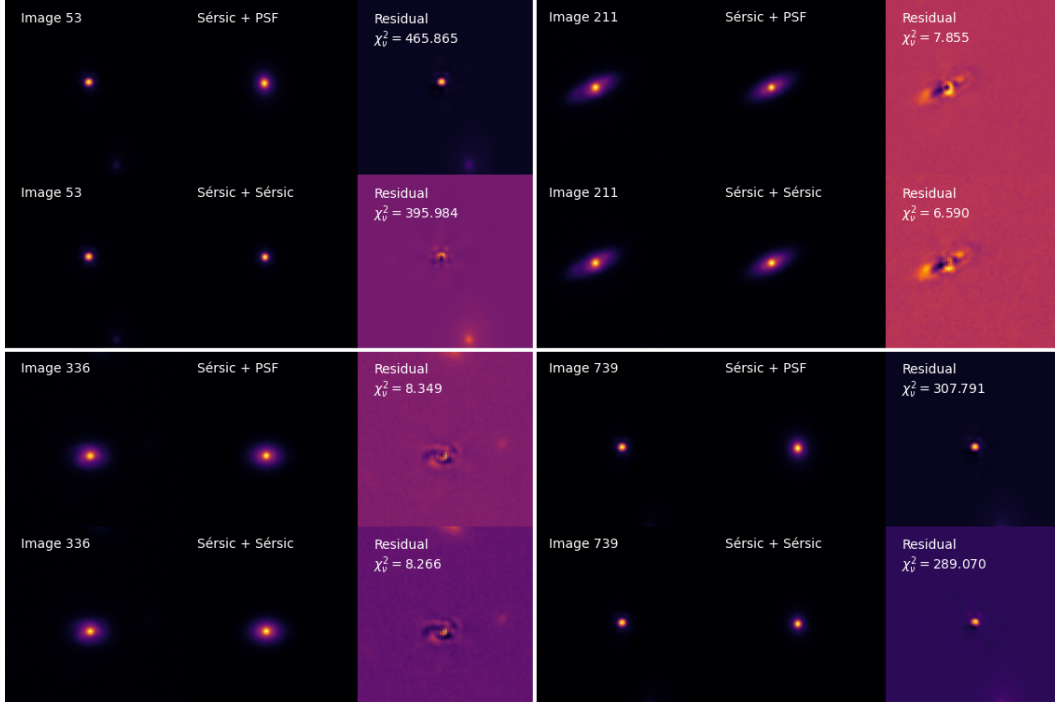


Figure 4.9 The fitting performed on the targets identified by the COSMOS-Web team as AGN. For each of the four targets there are six panels: from left to right we have the Target ID or Image Number, the model (either Sérsic + PSF or Sérsic + Sérsic) followed by the residual with χ^2_ν ; the top row has the fits for the Sérsic + PSF model and the bottom row has the Sérsic + Sérsic fit. Importantly, targets 53 and 739 are the same target, one initially identified with VLA and a second time in the NIR.

Chapter 5

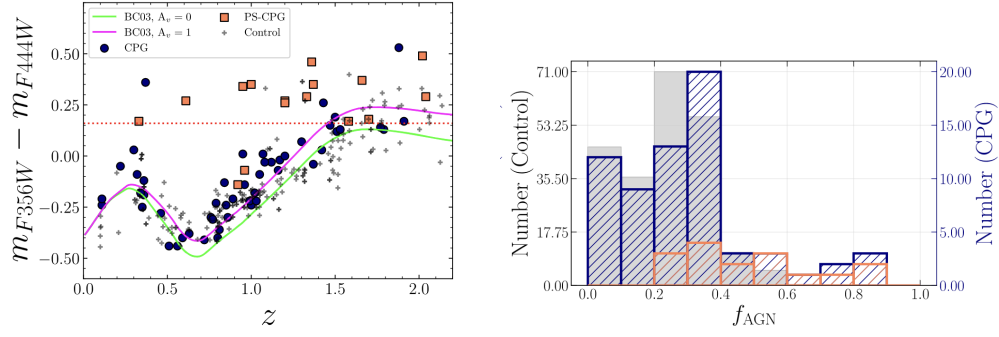
Discussion

5.1 Comparison with Ortiz et al. 2024

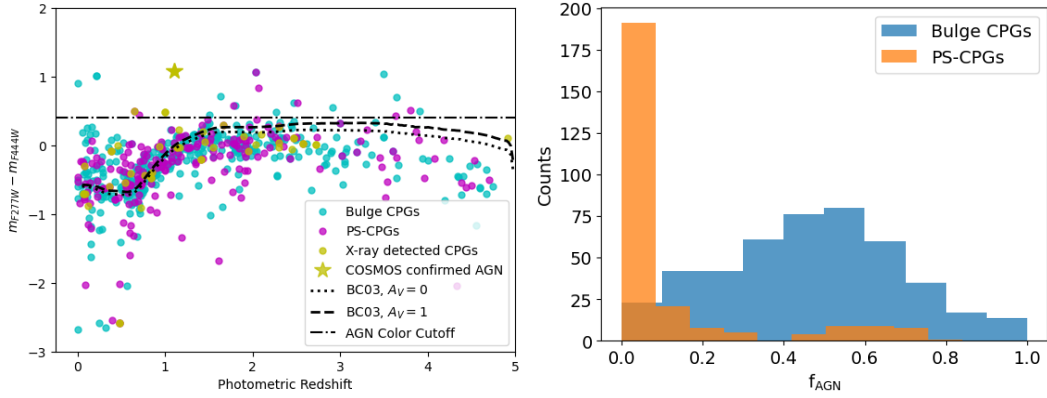
Compared to Ortiz et al. (2024), which found 16/66 or about 24% of the CPGs to have identifiable point sources, we found 123/385, about 32% of the CPGs we selected for having an unresolved point source. Ortiz et al. (2024) found 14 of the 16 PS-CPGs to be AGN based on their $m_{F356W} - m_{F444W}$ color, which makes up almost 90% of their PS-CPG sample, as we can see in Figure 5.1a. In this work, by comparison, we identified only 3 visually selected targets out of 123 to be AGN based on their $m_{F277W} - m_{F444W}$ color, which is approximately 2.4%. The other difference between the results of Ortiz et al. (2024) and our work is that the AGN fraction of our data is extremely low for the color selected PS-CPGs, at 0.0001, compared to 0.47 of Ortiz et al. seen in Figure 5.1b. If we ignore the 50 of 123 with an AGN fraction near 0, we get an f_{AGN} of 0.64, higher than what we expected.

5.2 New Insights from the COSMOS-Web/CCP Sample

We found 256 PS-CPGs out of the total 727 CPGs, which is about 35% of the original sample we selected (by eye in the NIR and radio). In this work we identified only 13 targets out of 256 to be AGN based on their $m_{F277W} - m_{F444W}$ color, which is approximately 5%. The AGN fraction of our data is extremely



(a) Color selection criterion from Ortiz et al. (2024). (b) f_{AGN} distribution from Ortiz et al. (2024).



(c) Color selection criterion as explained in Figure 4.7. (d) AGN fraction as shown in Figure 4.3.

Figure 5.1 (a) Color selection criterion from Ortiz et al. (2024) for their CPG sample of 66. The color filters are different than (c). Any of the PS-CPGs that lay above the red dotted line are considered AGN. (b) AGN fraction from Ortiz et al. (2024) showing the distributions for the control group in gray, the Bulge CPGs in blue and the PS-CPGs in orange. (c) Color selection described in this work. (d) AGN fraction described in this work. Compared to (d) the distribution of AGN fraction from 0.4 to 0.8 in our sample matches the shape of the distribution here in orange.

low for the color selected PS-CPGs, at 0.007. If we ignore the 197 out of 256 with an AGN fraction near 0, we can get an f_{AGN} of 0.45, which is about what we expected.

We also analyzed our CPGs in the context of multi-wavelength tracers, specif-

ically radio and x-ray. This analysis revealed an additional 30 galaxies that are very likely to be AGN because they were detected in three separate wavelengths that are strong markers of AGN. The biggest limiting factor for these identifications is the x-ray detections. There are 429 targets detected in the IR and radio by VLA and JWST COSMOS-Web, while only 60 total x-ray detections. We lose a lot of possible targets by requiring x-ray detection for AGN confirmation. The median redshift ($z = 1.12$) of our targets makes it so that x-ray detectors would have trouble with some of our higher redshift targets, possibly contributing to a low number of multi-wavelength detections.

5.3 Fitting Challenges

We were unable to fit 181 of the 908 targets, and they all presented the same error. We removed these 181 from the fitting catalog and worked only with the remaining 727 CPGs in this thesis. The fitting fails when `pysersic` cannot properly identify the target. This can happen for a variety of reasons: source confusion or incomplete survey coverage, as we can see in 5.2. Targets such as 12 and 102 from Figure 5.2 are confused sources that would require more specific masks around the target of interest.

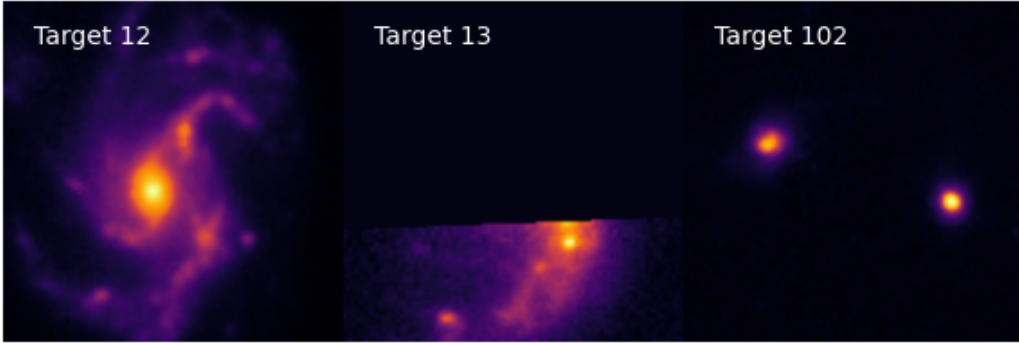


Figure 5.2 Examples of target galaxies that fail to be fit by `pysersic`. This shows the limitations of `pysersic` and the possible loss of many viable targets. Although, targets like 13 that are incomplete are correctly removed from the dataset in this case.

Additionally, there were a number of CPGs that `pysersic` fit with a very low AGN fraction. We would have to conduct further exploration to determine the exact problem with the fitting. Specifically because it only occurs with the Sérsic + PSF profile fit. We could also compare the morphological fitting

of `pyesric` to another fitting program `Galfit` to see if the issue is limited to `pyesric`'s f_{AGN} calculations or is merely a symptom of a broader issue with `pyesric`. There is also the chance it could be the result of the default priors set for the Sérsic + PSF profile.

Our closer analysis of the 0 flux targets reveals a combination of helpful and unhelpful results: sometimes there is no target within the FOV for `pyesric` to fit; other times the target appears perfectly point-like but is not fit with a high AGN fraction. A likely scenario is that these sources with 0 AGN amplitude might be described best by a single Sérsic profile that has little AGN contribution.

Chapter 6

Conclusion

6.1 Summary

We used `pysersic` to fit a Sérsic profile with a point-source component as well as a double Sérsic profile to 908 galaxies, 727 of which had measurable point-source components. We compared the flux ratios and the results differ in some cases with those of [Ortiz et al. \(2024\)](#). Of the sources that preferred the Sérsic with point-source profile fit, the flux from the central point was generally quite low. Core classification suggests that there are 256 PS-CPGs in our sample; 13 of which we expect to host IR-luminous AGN. We summarize our most important conclusions below:

1. We identify 3 of 123 targets with unresolved PSFs in the NIR as AGN via their $m_{F277W} - m_{F444W}$ colors;
2. The median f_{ps} value for the 3 color-identified AGN is 0.0001;
3. If we ignore all 50 PS-CPGs of the 123 with f_{AGN} below 0.1, the median is 0.63; much higher than the 0.47 that [Ortiz et al. \(2024\)](#) found;
4. The 256 PS-CPGs are split 75 NIRCam to 181 VLA detections;
5. We identify 13 of 256 NIR and radio selected targets as AGN via their $m_{F277W} - m_{F444W}$ colors;

6. The median f_{ps} value for the 13 color-identified AGN is 0.007;
7. If we ignore all 197 PS-CPGs with f_{AGN} below 0.1, the median is 0.45;
8. We detect 30 of 702 COSMOS-Web position-matched NIR sources in three separate wavelengths: NIR, radio, and X-ray. These are likely to be AGN.

6.2 Limitations of our Method

Our method of identifying galaxies with a point source element relied on by-eye measurements and estimations instead of utilizing a program like **SExtractor** to identify point sources as [Ortiz et al. \(2024\)](#). We could also improve calculation of the χ^2_ν values with either an improvement on the masking or a better method of generating a sub-frame. To improve the masking, we would need to not only mask the background but other nearby targets within the cutout scope. The other option is improving the sub-frame generation. Ideally the sub-frame would not zoom in on the middle 1/3 of the image but would instead focus on the 1/3 of the image that kept the target centered in the sub-frame. We discuss the χ^2_ν calculations further in Appendix A.

We lose a lot of sources due to failures during fitting, as we discuss in [5.3](#). With a more reliable method of masking out unwanted sources, the full-frame cutouts could cover a wider FOV and ideally, we would get use some of those lost viable sources such as target 12 in Figure [5.2](#).

We cannot accurately compare the AGN fraction of the color selected AGN to [Ortiz et al. \(2024\)](#) since many of our PS-CPGs were incorrectly fit to have a near zero AGN fraction. We believe that were this error corrected, the f_{AGN} of the 13 color selected AGN would be nearer to that of [Ortiz et al. \(2024\)](#). That being said, we did not detect nearly the same number of color selected AGN as Ortiz et al., which could be due to the differing filters or the cutoff point.

6.3 Insights into Obscured AGN

Our method provides the opportunity to identify obscured AGN without as much data in multiple wavelengths. This could be particularly useful with existing deep-field data, making it possible to find additional obscured AGN.

Learning more about dusty AGN is important because there are still many questions about the formation and possible coevolution of SMBH and galaxies in the early universe. We hope that with this improved method of identifying unresolved IR-luminous AGN we can discover new populations of AGN that might have otherwise been missed and can broaden our understanding of AGN populations as a whole. This should allow us to gain a better understanding of the active phase of AGN and how that plays into galaxy and star formation. How did the SMBH initially get to the center of the galaxy? Did they form alongside it? Or did they get pulled into the center of the galaxy through a merger or by gravity? We know that AGN feedback has the potential to effect space on the kpc or Mpc scale, although we don't yet fully understand how these functions are powered and how they influence the average galaxy.

6.4 Next Steps

With all the available data from the COSMOS-Web field many more investigations of the selected targets could be possible. In this work, we examined specific morphological parameters with `pyrsic` and a fraction of the data available from the COSMOS-Web 2025 DR1 catalog. Other data, including star formation rates and mass estimations, could be used for comparisons and further confirmation of the structure of these objects. Using spectral energy distribution analysis and spectroscopic redshift could also enhance our analysis. Even with their limitations, our methods offer a new way to search for dust obscured IR-luminous AGN in deep-field NIR imaging.

Part III

Appendices

Appendix A

Appendix

A.1 Statmorph

In a previous incarnation of this project we used `statmorph`, an affiliated package of `astropy`, designed for non-parametric morphological diagnostics of galaxy images. After fitting targets and sorting through the fits we found that, for the purposes of this project and for this set of data, it did not perform satisfactory fitting. In many cases, fitting the wrong target (if there was a secondary background object) or creating a fit that was clearly off. The program was often unable to differentiate between two targets if the separation was low. The biggest reason we chose to switch to `pysersic` from `Statmorph` was the profile fits. Both are able to perform Sérsic and double-Sérsic fits but there was not, as far as we could tell, a way to perform a Sérsic profile fit with a PSF component using `Statmorph`. An example of a `Statmorph` double-Sérsic fit can be seen in [A.1](#).

A.2 Calculating χ^2

`pysersic` does not compute a goodness-of-fit value, such as a reduced χ^2 value for each fit. Instead it produces an array of loss values using whichever profile was specified during fitting. A second loss value is generated during the fitting process. These values did not contribute to the analysis of the results in this work, although a loss function is required for `pysersic` when fitting. Here I use the Student T loss function as used in the `pysersic` tutorials. This

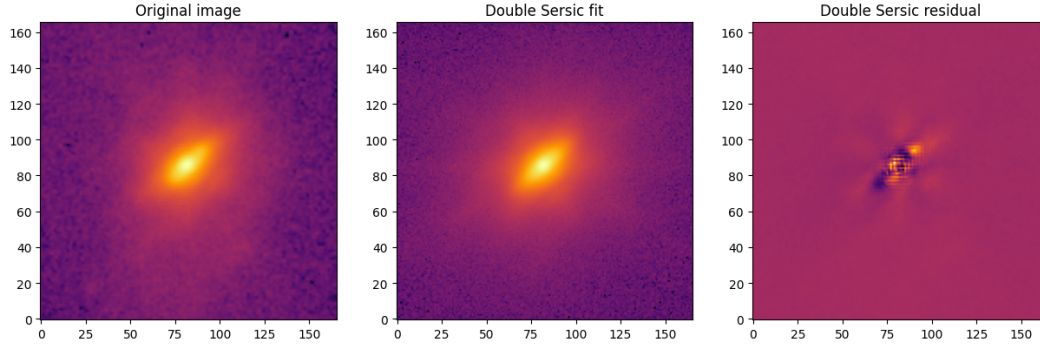


Figure A.1 An example of a target fit with a double-Sérsic profile fit with **Statmorph** the reduced χ^2 for this fit was 5.17.

function has "fatter tails" than a gaussian or χ^2 loss function and so is more resistant to outliers ([Pasha & Miller, 2023](#)).

When we decided to calculate the χ^2 values, we tried multiple methods. Calculating the χ^2 statistic for the entire $5'' \times 5''$ image resulted in values that were sometimes in the thousands. The first method we tried to remedy this problem was to recalculate the fit on a zoomed in portion of the original fit. In this case, it meant not changing the fit but looking at the middle section of the image for the calculation of the χ^2 statistic. This method turned out to be the most reliable and is the method used in the analysis. The second method we tried involved taking the center section of the image and refitting it with both profiles. During testing this appeared to be the better solution with model fits matching the targets much better. In practice, after rerunning the pipeline, we found that this method was actually much worse, as can be seen in [Figure A.4](#). This methodology was also worse in the cases where the target was cut out of the sub-frame because **pysersic** would force a fit to the region, as we can see in [Figure A.2](#). Or in other cases **pysersic** would pick an entire new target to perform fitting on as we can see in [Figure A.3](#)

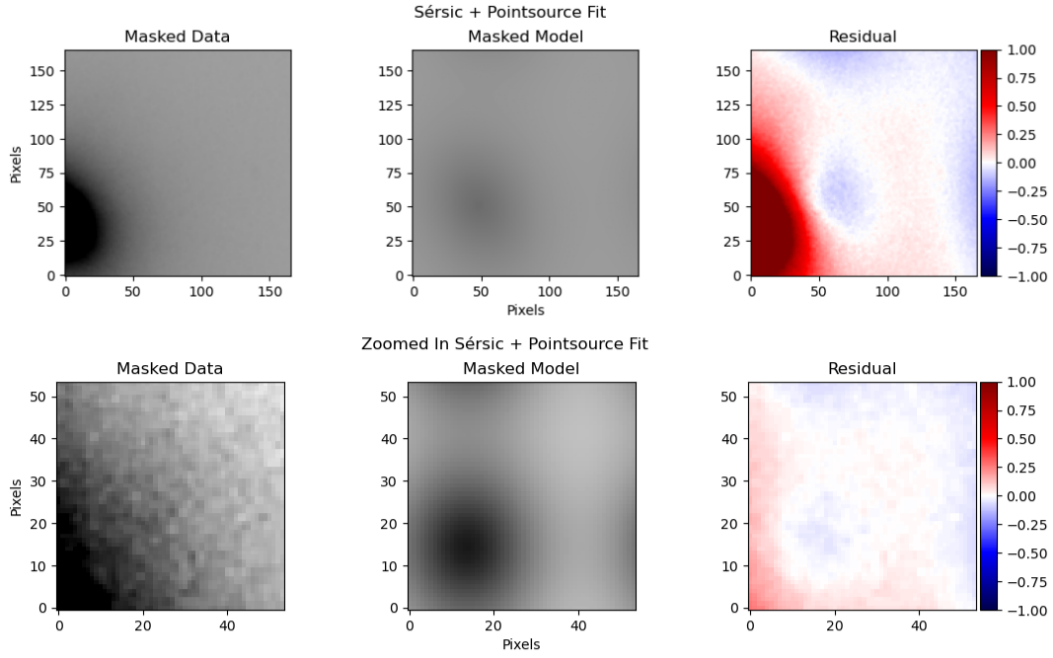


Figure A.2 The top row shows the Sérsic + PSF fit for the full frame FOV and the bottom row shows the Sérsic + PSF fit for the sub-frame. From left to right the images are the masked data, the masked model (Sérsic + PSF), and the residual. The Sérsic+PSF profile fit of a target that does not fit properly within the chosen FOV. This demonstrates how `pysersic` will sometimes fit a target even if one doesn't exist, creating a false "improved" fit.

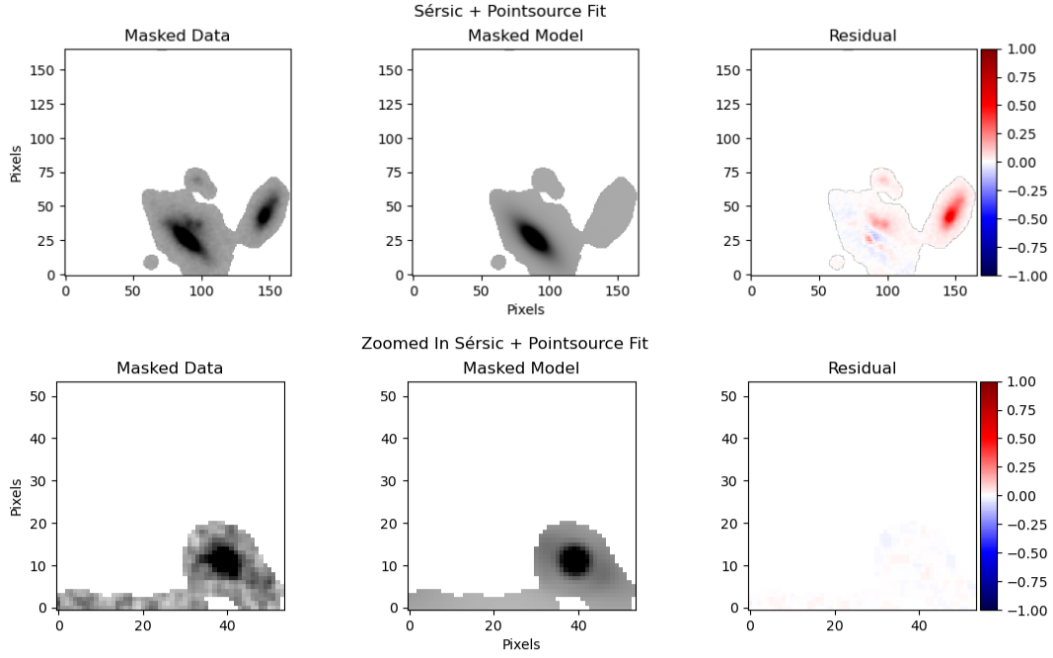


Figure A.3 The top row shows the Sérsic + PSF fit for the full frame FOV and the bottom row shows the Sérsic + PSF fit for the sub-frame. From left to right the images are the masked data, the masked model (Sérsic + PSF), and the residual. The Sérsic+PSF profile fit of a target that does not fit properly within the chosen FOV. This demonstrates how `pysersic` will pick a new target if the original is out of view, creating an unreliable new model fit that no longer contributes to the data.

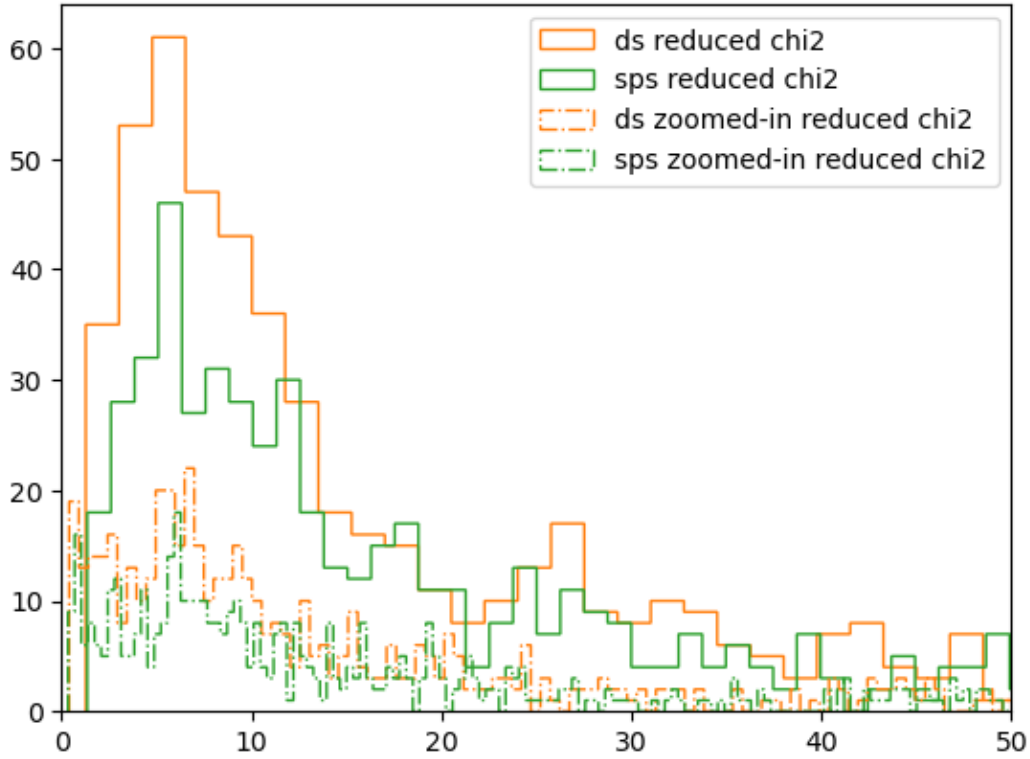


Figure A.4 The comparison of the distribution of reduced χ^2 values after re-fitting the sub-frame images. The dashed lines represent the distribution of the new reduced χ^2 statistics after refitting the sub-frame. The solid line represents the reduced χ^2 statistics from fitting the full-frame. The distributions of reduced χ^2 after refitting are less useful.

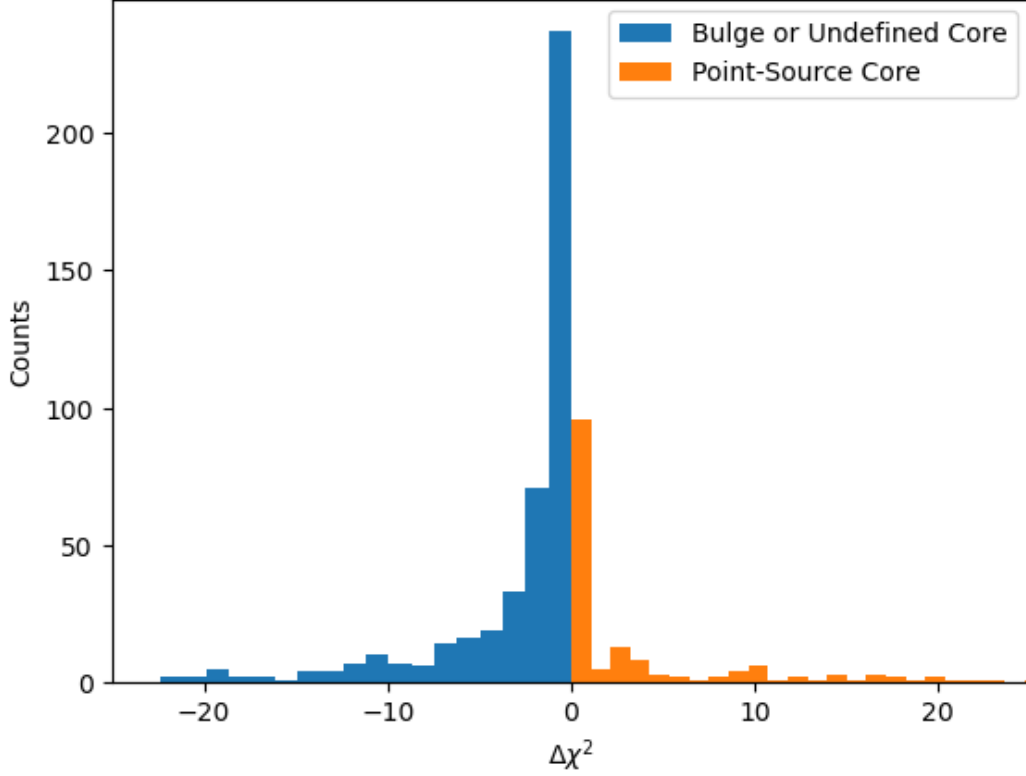


Figure A.5 Based on Figure 4 of (Ortiz et al., 2024). A Histogram of the χ^2 goodness-of-fit statistic between the double-Sérsic and Sérsic+PSF fits after refitting the sub-frame. $\Delta\chi^2 = \chi^2_{double-Sérsic} - \chi^2_{Sérsic+PSF}$ indicating a predilection for either a bulge or point-source core. Compared to Figure 4.1 there are far fewer point-source core classifications due to the less useful calculation of reduced χ^2 , as we discuss in Figure A.4.

Appendix B

Code and Files

B.1 Code

All code used for this project is stored on a Github [repository](#).

The code to complete these analyses relies on the following packages: `pysersic`, `STPSF`, `numpy`, `astropy`, `pandas`, `matplotlib.pyplot`, and `sys`

Tutorials can be found here: [STPSF](#) and [pysersic](#).

code for generating a $4.98'' \times 4.98''$ PSF and saving it. Appropriate changes can easily be made for different settings. This code is directly following tutorials within the STPSF documentation.

```
1 import stpsf
2
3 nrc = stpsf.NIRCam()
4 nrc.filter = 'F444W'
5 nrc.options['parity'] = 'even'
6 nrc.pixelscale = .03 # 30mas
7 psf = nrc.calc_psf( fov_arcsec = 4.98)
8 # returns an astropy.io.fits.HDUlist containing PSF and header
9
10 datadir = '/Users/brownscholar/Desktop/Education/College Course
           Work/Fourth Year/Thesis/Research/Coding/Data/'
11 psf.writeto(datadir + 'generated_psf_4.98as.fits', overwrite =
           True)
```

Code for running the pysersic pipeline and saving the results as a csv file:

```
1 import numpy as np
2 import pandas as pd
3 from astropy.io import fits
4 from pysersic.results import plot_image
5 from pysersic import check_input_data
6 from pysersic.priors import autoprior
7 from pysersic import FitSingle
8 from pysersic.loss import student_t_loss
9 from jax.random import PRNGKey # Need to use a seed to start
   jax's random number generation
10 from pysersic.results import plot_residual
11 import pysersic as py
12 import matplotlib.pyplot as plt
13 import jax.numpy as jnp
14 import sys
15 np.set_printoptions(threshold=sys.maxsize)
16
17 data_path = '/Users/brownscholar/Desktop/Education/College
   Course Work/Fourth Year/Thesis/Research/Coding/Data/'
18
19 def open_extract(im_num,n,psfn,data_dir):
20     '''
21     a function to open and extract the data from the set of
22     fits files needed
23     for morphological fitting by pysersic. which include: image
24     , mask, uncertainty
25     or error, and psf
26
27     PARAMETERS
28     im_num: indentifying number in image file name
29     n: index of the data you want to extract from the given
30     files
31     psfn: index of the data you want to extract from the given
32     psf file
33     data_dir: shared part of the file path to your data
34
35     RETURNS
36     the extracted data from all give files as well as the mask
37     data in two forms:
38     an inverted boolean array and the array data extracted in
39     its original form.
40     '''
41
42     im = fits.open(f'{data_dir}COSMOS-Web Cutouts/sci/
43 mosaic_nircam_f444w_COSMOS-Web_30mas_B4_v1.0_sci_small_5_{
44 im_num}.fits',allow_pickle=True)
```

```

37     mask = fits.open(f'{data_dir}COSMOS-Web Cutouts/segmaps/
detection_chi2pos_SWLW_B4_segmap_v1.3_small_5_{im_num}.fits
',allow_pickle=True)
38     sig = fits.open(f'{data_dir}COSMOS-Web Cutouts/err/
mosaic_nircam_f444w_COSMOS-Web_30mas_B4_v1.0_err_small_5_{
im_num}.fits',allow_pickle=True)
39     psf = fits.open(f'/{data_dir}generated_psf_4.98as.fits',
allow_pickle=True)
40
41     im_data = im[n].data
42     mask_data = np.array(mask[n].data)
43     mask_data_flip = np.invert(np.array(mask[n].data,dtype=bool
))
44     sig_data = sig[n].data
45     psf_data = psf[psfn].data
46     return im_data, mask_data_flip, sig_data, psf_data,
mask_data #sm_psf_data
47
48 def fitting (im_data, mask_data, sig_data, psf_data, prof_type:
str, loss_func=student_t_loss, sky = 'none'):
49     '''
50     Function to run fitting on a given fits file(s)
51
52     PARAMETERS
53     im_data: array - data array from given fits file
54     mask_data: boolean array - indicator of target to fit
55     sig_data: array - array of error or uncertainty values
56     psf_data: array - psf to calculate loss, same size as image
57     prof_type: string - which fitting profile is to be used on
the given target
58     loss_func: func - name of loss function from Pysersic to
run, default = student_t_loss
59     sky: string- sky type for fitting, default none
60
61     RETURNS
62     map_params: pysersic model fit of whole image
63     map_params_lim: pysersic model fit of the middle 1/3 of the
image
64     '''
65
66     prior = autoprior(image=im_data, profile_type=prof_type,
mask=mask_data, sky_type=sky)
67     fitter = FitSingle(data=im_data, rms=sig_data, mask=
mask_data, psf=psf_data, prior=prior, loss_func=loss_func)
68     map_params = fitter.find_MAP(rkey=PRNGKey(1000))
69
70     prior_lim = autoprior(image=im_data[58:112,58:112],
profile_type=prof_type, mask=mask_data[58:112,58:112],

```

```

sky_type = 'none')
71     fitter_lim = FitSingle(data=im_data[58:112,58:112],rms=
sig_data[58:112,58:112], mask=mask_data[58:112,58:112],psf=
psf_data[58:112,58:112], prior=prior_lim,loss_func=
student_t_loss)
72     map_params_lim = fitter_lim.find_MAP(rkey=PRNGKey(1000))
73
74     return map_params, map_params_lim
75
76 def running(num,data_dir):
77     '''
78     a function to run the fitting function and sort the output
into a dataframe to be worked with elsewhere
79
80     PARAMETERS
81     num: total number of files to attempt to perform fitting on
82     data_dir: shared part of file path to data
83
84     RETURNS
85     sps_results: the results of fitting a S rsic+PSF profile
to the data
86     ds_results: the results of fitting a double-S rsic profile
to the data
87
88     '''
89
90     sps_results = pd.DataFrame({'image_num':[],'flux':[],'xc'
: [],
91                                'yc':[],'r_eff':[],'ellip':[],
92                                'theta':[], 'n':[],'f_psf':[],
93                                'model':None,'res':None, '
red_chi2':[],
94                                'zoom_red_chi2':[],'chi2_ratio'
: []})
95
96     ds_results = pd.DataFrame({'image_num':[],'flux':[],'xc'
: [],
97                                'yc':[], 'f_1':[],'theta':[],
98                                'r_eff_1':[],'r_eff_2':[],
99                                'ellip_1':[],'ellip_2':[],
100                                'n_1':[],'n_2':[],'model':None, '
res':None,
101                                'red_chi2':[],'zoom_red_chi2'
: [],'chi2_ratio': []})
102
103
104     for i in range(num):
105         try:

```

```

106         print(f'\nFitting: {i}')
107         sci_data,mask_data_flip,err_data,psf_data,mask_data
= open_extract(i,0,3,data_dir)
108
109         spsmap = fitting(im_data=sci_data,mask_data=
mask_data_flip,
110                             sig_data=err_data,psf_data=
psf_data,
111                             prof_type='
sersic_pointsource')
112
113         dsmmap = fitting(im_data=sci_data,mask_data=
mask_data_flip,
114                             sig_data=err_data,psf_data=
psf_data,
115                             prof_type='doublesersic')
116
117         sps_1d_mod = np.ravel(spsmap['model'])
118         ds_1d_mod = np.ravel(dsmmap['model'])
119
120         sps_res = sci_data-spsmap['model']
121         ds_res = sci_data-dsmmap['model']
122
123         sps_1d_res = np.ravel(sps_res)
124         ds_1d_res = np.ravel(ds_res)
125
126         lim_N_eff = np.count_nonzero(mask_data
[58:112,58:112])
127         sps_dof = 8
128         ds_dof = 11
129         lim_sps_red_chi2 = (np.sum(((np.ma.array(sps_res
[58:112,58:112], mask=mask_data_flip[58:112,58:112])) / np.
ma.array(err_data[58:112,58:112], mask=mask_data_flip
[58:112,58:112]))**2))/ (lim_N_eff - sps_dof)
130         lim_ds_red_chi2 = (np.sum(((np.ma.array(ds_res
[58:112,58:112], mask=mask_data_flip[58:112,58:112])) / np.
ma.array(err_data[58:112,58:112], mask=mask_data_flip
[58:112,58:112]))**2))/ (lim_N_eff - ds_dof)
131
132
133         N_eff = np.count_nonzero(mask_data)
134         sps_red_chi2 = (np.sum(((np.ma.array(sps_res, mask=
mask_data_flip)) / np.ma.array(err_data, mask=
mask_data_flip)**2))/ (N_eff - sps_dof)
135         ds_red_chi2 = (np.sum(((np.ma.array(ds_res, mask=
mask_data_flip)) / np.ma.array(err_data, mask=
mask_data_flip)**2))/ (N_eff - ds_dof)
136

```

```

137     sps_results.loc[i, 'image_num'] = i
138     sps_results.loc[i, 'flux'] = spsmap['flux']
139     sps_results.loc[i, 'xc'] = spsmap['xc']
140     sps_results.loc[i, 'yc'] = spsmap['yc']
141     sps_results.loc[i, 'r_eff'] = spsmap['r_eff']
142     sps_results.loc[i, 'ellip'] = spsmap['ellip']
143     sps_results.loc[i, 'theta'] = spsmap['theta']
144     sps_results.loc[i, 'n'] = spsmap['n']
145     sps_results.loc[i, 'f_ps'] = spsmap['f_ps']
146     sps_results.at[i, 'model'] = sps_1d_mod
147     sps_results.at[i, 'res'] = sps_1d_res
148     sps_results.loc[i, 'red_chi2'] = sps_red_chi2
149     sps_results.loc[i, 'zoom_red_chi2'] =
lim_sps_red_chi2
150     sps_results.loc[i, 'chi2_ratio'] = lim_sps_red_chi2/
sps_red_chi2
151
152     ds_results.loc[i, 'image_num'] = i
153     ds_results.loc[i, 'flux'] = dsmap['flux']
154     ds_results.loc[i, 'xc'] = dsmap['xc']
155     ds_results.loc[i, 'yc'] = dsmap['yc']
156     ds_results.loc[i, 'f_1'] = dsmap['f_1']
157     ds_results.loc[i, 'r_eff_1'] = dsmap['r_eff_1']
158     ds_results.loc[i, 'r_eff_2'] = dsmap['r_eff_2']
159     ds_results.loc[i, 'ellip_1'] = dsmap['ellip_1']
160     ds_results.loc[i, 'ellip_2'] = dsmap['ellip_2']
161     ds_results.loc[i, 'theta'] = dsmap['theta']
162     ds_results.loc[i, 'n_1'] = dsmap['n_1']
163     ds_results.loc[i, 'n_2'] = dsmap['n_2']
164     ds_results.at[i, 'model'] = ds_1d_mod
165     ds_results.at[i, 'res'] = ds_1d_res
166     ds_results.loc[i, 'red_chi2'] = ds_red_chi2
167     ds_results.loc[i, 'zoom_red_chi2'] = lim_ds_red_chi2
168     ds_results.loc[i, 'chi2_ratio'] = lim_ds_red_chi2/
ds_red_chi2
169
170     except:
171         print(f'Error at: {i}!')
172     sps_results = sps_results.reset_index()
173     ds_results = ds_results.reset_index()
174
175     ds_results['image_num'] = ds_results['index']
176     sps_results['image_num'] = ds_results['index']
177
178     return sps_results.drop('index', axis=1), ds_results.drop('
index', axis=1)
179
180 spsresults, dsresults = running(908, data_path)

```



```
181
182 spsresults.to_csv('Data/pysersic_results/sps_results_maskedchi2
    .csv')
183 dsresults.to_csv('Data/pysersic_results/ds_results_maskedchi2.
    csv')
```

Acronyms

AGN active galactic nuclei. [1](#), [2](#), [68](#)

BH black hole. [14](#), [68](#)

Bulge-CPG Bulge Central Point-Source Galaxies. [68](#)

CPG Central Point-Source Galaxies. [68](#)

FOV field of view. [28](#), [68](#)

IR infrared. [4](#), [68](#)

JWST James Webb Space Telescope. [17](#), [68](#)

MBH massive black holes. [13](#), [68](#)

NIR near-infrared. [26](#), [68](#)

NIRCam Near-Infrared Camera. [27](#), [68](#)

PS-CPG Point-Source Central Point-Source Galaxies. [68](#)

PSF point spread function. [25](#), [26](#), [68](#)

QSG quasi-stellar galaxies. [4](#), [68](#)

QSO quasi-stellar objects. [7](#), [68](#)

SMBH supermassive black hole. [17](#), [68](#)

VLA Very Large Array. [17](#), [68](#)

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